# IDEAL GASES

## 15.1 The mole

Candidates should be able to:

1 understand that amount of substance is an SI base quantity with the base unit mol

2 use molar quantities where one mole of any substance is the amount containing a number of particles of that substance equal to the Avogadro constant $N_A$

## 15.2 Equation of state

Candidates should be able to:

1 understand that a gas obeying $pV \propto T$, where $T$ is the thermodynamic temperature, is known as an ideal gas

2 recall and use the equation of state for an ideal gas expressed as $pV = nRT$, where $n$ = amount of substance (number of moles) and as $pV = NkT$, where $N$ = number of molecules

3 recall that the Boltzmann constant $k$ is given by $k = R / N_A$

## 15.3 Kinetic theory of gases

Candidates should be able to:

1 state the basic assumptions of the kinetic theory of gases

2 explain how molecular movement causes the pressure exerted by a gas and derive and use the relationship $pV = \frac{1}{3}Nm\langle c^2\rangle$, where $\langle c^2\rangle$ is the mean-square speed (a simple model considering one-dimensional collisions and then extending to three dimensions using $\frac{1}{3}\langle c^2\rangle = \langle c_x^2\rangle$ is sufficient)

3 understand that the root-mean-square speed $c_{r.m.s.}$ is given by $\sqrt{\langle c^2\rangle}$

4 compare $pV = \frac{1}{3}Nm\langle c^2\rangle$ with $pV = NkT$ to deduce that the average translational kinetic energy of a molecule is $\frac{3}{2}kT$

**1. O/N/07/Q2**

**(a)** An amount of 1.00 mol of Helium-4 gas is contained in a cylinder at a pressure of $1.02 \times 10^5$ Pa and a temperature of 27 °C.

**(i)** Calculate the volume of gas in the cylinder.

volume = ............................................ $m^3$ [2]

**(ii)** Hence show that the average separation of gas atoms in the cylinder is approximately $3.4 \times 10^{-9}$ m.

[2]

**(b)** Calculate

**(i)** the gravitational force between two Helium-4 atoms that are separated by a distance of $3.4 \times 10^{-9}$ m,

force = ............................................... N [3]

(ii) the ratio

$$\frac{\text{weight of a Helium-4 atom}}{\text{gravitational force between two Helium-4 atoms with separation } 3.4 \times 10^{-9}\,\text{m}}.$$

ratio = ....................................................[2]

**(c)** Comment on your answer to **(b)(ii)** with reference to one of the assumptions of the kinetic theory of gases.

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**2. M/J/08/Q2**

**(a)** Explain qualitatively how molecular movement causes the pressure exerted by a gas.

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**(b)** The density of neon gas at a temperature of 273 K and a pressure of $1.02 \times 10^5$ Pa is $0.900\,\text{kg}\,\text{m}^{-3}$. Neon may be assumed to be an ideal gas.

Calculate the root-mean-square (r.m.s.) speed of neon atoms at

**(i)** 273 K,

speed = ............................................ $\text{m}\,\text{s}^{-1}$ [3]

**(ii)** 546 K.

speed = ............................................ $\text{m}\,\text{s}^{-1}$ [2]

**(c)** The calculations in **(b)** are based on the density for neon being 0.900 kg $m^{-3}$.
Suggest the effect, if any, on the root-mean-square speed of changing the density at constant temperature.

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**3. *O/N/09/41/Q2***

An ideal gas occupies a container of volume $4.5 \times 10^3\,cm^3$ at a pressure of $2.5 \times 10^5\,Pa$ and a temperature of 290 K.

**(a)** Show that the number of atoms of gas in the container is $2.8 \times 10^{23}$.

[2]

**(b)** Atoms of a real gas each have a diameter of $1.2 \times 10^{-10}\,m$.

**(i)** Estimate the volume occupied by $2.8 \times 10^{23}$ atoms of this gas.

volume = .......................................... $m^3$ [2]

**(ii)** By reference to your answer in **(i)**, suggest whether the real gas does approximate to an ideal gas.

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.................................................................................................................... [2]

**4. M/J/10/41/Q2**

**(a)** Some gas, initially at a temperature of 27.2 °C, is heated so that its temperature rises to 38.8 °C.
Calculate, in kelvin, to an appropriate number of decimal places,

**(i)** the initial temperature of the gas,

initial temperature = .............................................. K [2]

**(ii)** the rise in temperature.

rise in temperature = .............................................. K [1]

**(b)** The pressure $p$ of an ideal gas is given by the expression

$$p = \frac{1}{3}\rho \langle c^2 \rangle$$

where $\rho$ is the density of the gas.

**(i)** State the meaning of the symbol $\langle c^2 \rangle$.

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**(ii)** Use the expression to show that the mean kinetic energy $\langle E_K \rangle$ of the atoms of an ideal gas is given by the expression

$$\langle E_K \rangle = \frac{3}{2}kT.$$

Explain any symbols that you use.

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**(c)** Helium-4 may be assumed to behave as an ideal gas.
A cylinder has a constant volume of $7.8 \times 10^3\,\text{cm}^3$ and contains helium-4 gas at a pressure of $2.1 \times 10^7\,\text{Pa}$ and at a temperature of 290 K.

Calculate, for the helium gas,

**(i)** the amount of gas,

amount = .......................................... mol [2]

**(ii)** the mean kinetic energy of the atoms,

mean kinetic energy = .............................................. J [2]

**(iii)** the total internal energy.

internal energy = .............................................. J [3]

**5. O/N/10/43/Q2**

**(a)** State the basic assumptions of the kinetic theory of gases.

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**(b)** Use equations for the pressure of an ideal gas to deduce that the average translational kinetic energy $<E_K>$ of a molecule of an ideal gas is given by the expression

$$<E_K> = \frac{3}{2}\frac{R}{N_A}T$$

where $R$ is the molar gas constant, $N_A$ is the Avogadro constant and $T$ is the thermodynamic temperature of the gas.

[3]

**(c)** A deuterium nucleus $^2_1H$ and a proton collide. A nuclear reaction occurs, represented by the equation

$$^2_1H + {}^1_1p \longrightarrow {}^3_2He + \gamma.$$

**(i)** State and explain whether the reaction represents nuclear fission or nuclear fusion.

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**(ii)** For the reaction to occur, the minimum total kinetic energy of the deuterium nucleus and the proton is $2.4 \times 10^{-14}$ J.
Assuming that a sample of a mixture of deuterium nuclei and protons behaves as an ideal gas, calculate the temperature of the sample for this reaction to occur.

temperature = ............................................... K [3]

**(iii)** Suggest why the assumption made in **(ii)** may not be valid.

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***6. M/J/11/41/Q2***

**(a)** State what is meant by the *Avogadro constant* $N_A$.

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**(b)** A balloon is filled with helium gas at a pressure of $1.1 \times 10^5$ Pa and a temperature of 25 °C.
The balloon has a volume of $6.5 \times 10^4\,\text{cm}^3$.
Helium may be assumed to be an ideal gas.

Determine the number of gas atoms in the balloon.

number = ................................................... [4]

**7. M/J/11/42/Q2**

**(a)** State what is meant by a *mole*.

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**(b)** Two containers A and B are joined by a tube of negligible volume, as illustrated in Fig. 2.1.

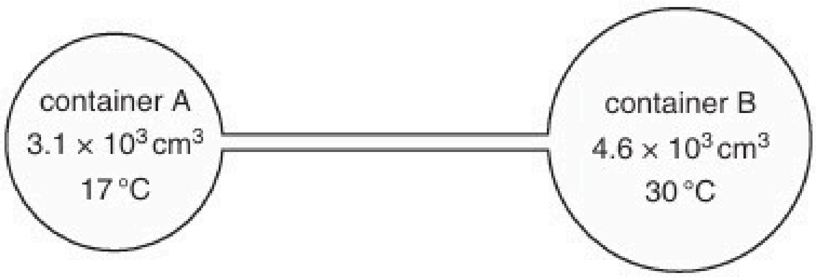


**Fig. 2.1**

The containers are filled with an ideal gas at a pressure of $2.3 \times 10^5$ Pa.
The gas in container A has volume $3.1 \times 10^3\,cm^3$ and is at a temperature of 17 °C.
The gas in container B has volume $4.6 \times 10^3\,cm^3$ and is at a temperature of 30 °C.

Calculate the total amount of gas, in mol, in the containers.

amount = ......................................... mol [4]

**8. O/N/11/41/Q2**

**(a)** One assumption of the kinetic theory of gases is that gas molecules behave as if they are hard, elastic identical spheres.

State two other assumptions of the kinetic theory of gases.

1. ..........................................................................................................................

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2. ..........................................................................................................................

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[2]

**(b)** Using the kinetic theory of gases, it can be shown that the product of the pressure $p$ and the volume $V$ of an ideal gas is given by the expression

$$pV = \tfrac{1}{3}Nm\langle c^2\rangle$$

where $m$ is the mass of a gas molecule.

**(i)** State the meaning of the symbol

**1.** $N$,

.......................................................................................................................... [1]

**2.** $\langle c^2\rangle$.

.......................................................................................................................... [1]

**(ii)** Use the expression to deduce that the mean kinetic energy $\langle E_K\rangle$ of a gas molecule at temperature $T$ is given by the equation

$$\langle E_K\rangle = \tfrac{3}{2}kT$$

where $k$ is a constant.

[2]

**(c)** **(i)** State what is meant by the *internal energy* of a substance.

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**(ii)** Use the equation in **(b)(ii)** to explain that, for an ideal gas, a change in internal energy $\Delta U$ is given by

$$\Delta U \propto \Delta T$$

where $\Delta T$ is the change in temperature of the gas.

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**9. O/N/11/43/Q1**

The planet Mars may be considered to be an isolated sphere of diameter $6.79 \times 10^{6}$ m with its mass of $6.42 \times 10^{23}$ kg concentrated at its centre.
A rock of mass 1.40 kg rests on the surface of Mars.

For this rock,

**(a)** **(i)** determine its weight,

weight = ............................................ N [3]

**(ii)** show that its gravitational potential energy is $-1.77 \times 10^{7}$ J.

[2]

**(b)** Use the information in **(a)(ii)** to determine the speed at which the rock must leave the surface of Mars so that it will escape the gravitational attraction of the planet.

speed = ........................................ $\text{m s}^{-1}$ [3]

**(c)** The mean translational kinetic energy $\langle E_K \rangle$ of a molecule of an ideal gas is given by the expression

$$\langle E_K \rangle = \frac{3}{2}kT$$

where $T$ is the thermodynamic temperature of the gas and $k$ is the Boltzmann constant.

**(i)** Determine the temperature at which the root-mean-square (r.m.s.) speed of hydrogen molecules is equal to the speed calculated in **(b)**.
Hydrogen may be assumed to be an ideal gas.
A molecule of hydrogen has a mass of 2 u.

temperature = ............................................. K [2]

**(ii)** State and explain one reason why hydrogen molecules may escape from Mars at temperatures below that calculated in **(i)**.

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**10. M/J/12/41/Q2**

**(a)** The kinetic theory of gases is based on some simplifying assumptions.
The molecules of the gas are assumed to behave as hard elastic identical spheres.
State the assumption about ideal gas molecules based on

**(i)** the nature of their movement,

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**(ii)** their volume.

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**(b)** A cube of volume $V$ contains $N$ molecules of an ideal gas. Each molecule has a component $c_X$ of velocity normal to one side S of the cube, as shown in Fig. 2.1.

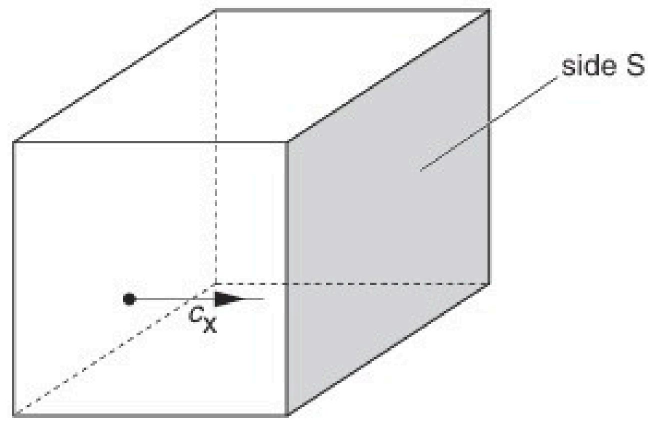


**Fig. 2.1**

The pressure $p$ of the gas due to the component $c_X$ of velocity is given by the expression

$$pV = Nmc_X^2$$

where $m$ is the mass of a molecule.

Explain how the expression leads to the relation

$$pV = \tfrac{1}{3}Nm\langle c^2\rangle$$

where $\langle c^2\rangle$ is the mean square speed of the molecules.

[3]

**(c)** The molecules of an ideal gas have a root-mean-square (r.m.s.) speed of $520\,\mathrm{m\,s^{-1}}$ at a temperature of 27 °C.

Calculate the r.m.s. speed of the molecules at a temperature of 100 °C.

r.m.s. speed = ........................................ $\mathrm{m\,s^{-1}}$ [3]

**11. O/N/12/43/Q1**

An ideal gas has volume $V$ and pressure $p$. For this gas, the product $pV$ is given by the expression

$$pV = \tfrac{1}{3}Nm\langle c^2\rangle$$

where $m$ is the mass of a molecule of the gas.

**(a)** State the meaning of the symbol

**(i)** $N$,

.......................................................................................................................... [1]

**(ii)** $\langle c^2\rangle$.

.......................................................................................................................... [1]

**(b)** A gas cylinder of volume $2.1 \times 10^4\,\text{cm}^3$ contains helium-4 gas at pressure $6.1 \times 10^5$ Pa and temperature 12 °C. Helium-4 may be assumed to be an ideal gas.

**(i)** Determine, for the helium gas,

**1.** the amount, in mol,

amount = .......................................... mol [3]

**2.** the number of atoms.

number = ................................................... [2]

**(ii)** Calculate the root-mean-square (r.m.s.) speed of the helium atoms.

r.m.s. speed = ....................................... $m\,s^{-1}$ [3]

**12. *M/J/13/41/Q2***

**(a)** State what is meant by an *ideal gas*.

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**(b)** Two cylinders A and B are connected by a tube of negligible volume, as shown in Fig. 2.1.

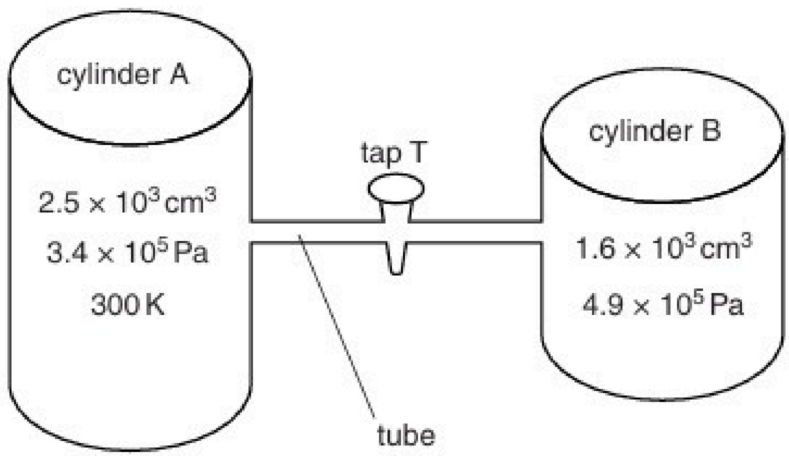


**Fig. 2.1**

Initially, tap T is closed. The cylinders contain an ideal gas at different pressures.

**(i)** Cylinder A has a constant volume of $2.5 \times 10^3\,\text{cm}^3$ and contains gas at pressure $3.4 \times 10^5\,\text{Pa}$ and temperature 300 K.

Show that cylinder A contains 0.34 mol of gas.

[1]

**(ii)** Cylinder B has a constant volume of $1.6 \times 10^3\,cm^3$ and contains 0.20 mol of gas. When tap T is opened, the pressure of the gas in both cylinders is $3.9 \times 10^5\,Pa$. No thermal energy enters or leaves the gas.

Determine the final temperature of the gas.

temperature = ................................................ K [2]

**(c)** By reference to work done and change in internal energy, suggest why the temperature of the gas in cylinder A has changed.

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***13. M/J/13/42/Q2***

**(a)** The volume of an ideal gas in a cylinder is $1.80 \times 10^{-3}\,m^3$ at a pressure of $2.60 \times 10^5\,Pa$ and a temperature of 297 K, as illustrated in Fig. 2.1.

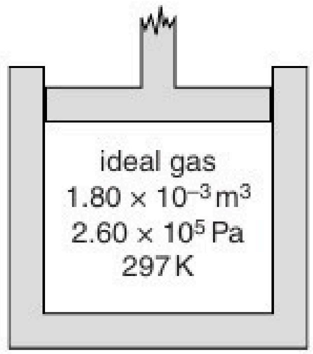


**Fig. 2.1**

The thermal energy required to raise the temperature by 1.00 K of 1.00 mol of the gas at constant volume is 12.5 J.

The gas is heated at constant volume such that the internal energy of the gas increases by 95.0 J.

**(i)** Calculate

**1.** the amount of gas, in mol, in the cylinder,

amount = ............................................ mol [2]

**2.** the rise in temperature of the gas.

temperature rise = ............................................ K [2]

**(ii)** Use your answer in **(i) part 2** to show that the final pressure of the gas in the cylinder is $2.95 \times 10^5$ Pa.

[1]

**(b)** The gas is now allowed to expand. No thermal energy enters or leaves the gas. The gas does 120 J of work when expanding against the external pressure.

State and explain whether the final temperature of the gas is above or below 297 K.

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**14. O/N/13/41/Q2**

The product of the pressure $p$ and the volume $V$ of an ideal gas is given by the expression

$$pV = \frac{1}{3}Nm\langle c^2\rangle$$

where $m$ is the mass of one molecule of the gas.

**(a)** State the meaning of the symbol

**(i)** $N$,

............................................................................................................................ [1]

**(ii)** $\langle c^2\rangle$.

............................................................................................................................ [1]

**(b)** The product $pV$ is also given by the expression

$$pV = NkT.$$

Deduce an expression, in terms of the Boltzmann constant $k$ and the thermodynamic temperature $T$, for the mean kinetic energy of a molecule of the ideal gas.

[2]

**(c)** A cylinder contains 1.0 mol of an ideal gas.

**(i)** The volume of the cylinder is constant.
Calculate the energy required to raise the temperature of the gas by 1.0 kelvin.

energy = ................................................ J [2]

**(ii)** The volume of the cylinder is now allowed to increase so that the gas remains at constant pressure when it is heated.
Explain whether the energy required to raise the temperature of the gas by 1.0 kelvin is now different from your answer in **(i)**.

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**15. M/J/14/41/Q2**

**(a)** Explain what is meant by the Avogadro constant.

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**(b)** Argon-40 ($^{40}_{18}Ar$) may be assumed to be an ideal gas.
A mass of 3.2 g of argon-40 has a volume of 210 $cm^3$ at a temperature of 37 °C.

Determine, for this mass of argon-40 gas,

**(i)** the amount, in mol,

amount = .................................................. mol [1]

**(ii)** the pressure,

pressure = ...................................................... Pa [2]

**(iii)** the root-mean-square (r.m.s.) speed of an argon atom.

r.m.s. speed = ................................................ $m\,s^{-1}$ [3]

**16. M/J/14/42/Q2**

A constant mass of an ideal gas has a volume of $3.49 \times 10^3\,\text{cm}^3$ at a temperature of 21.0 °C. When the gas is heated, 565 J of thermal energy causes it to expand to a volume of $3.87 \times 10^3\,\text{cm}^3$ at 53.0 °C. This is illustrated in Fig. 2.1.


**Fig. 2.1**

**(a)** Show that the initial and final pressures of the gas are equal.

[2]

**(b)** The pressure of the gas is $4.20 \times 10^5$ Pa.

For this heating of the gas,

**(i)** calculate the work done by the gas,

work done = ........................................................ J [2]

**(ii)** use the first law of thermodynamics and your answer in **(i)** to determine the change in internal energy of the gas.

change in internal energy = ...................................................... J [2]

**(c)** Explain why the change in kinetic energy of the molecules of this ideal gas is equal to the change in internal energy.

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**17. O/N/14/41/Q3**

**(a)** State what is meant by an *ideal* gas.

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**(b)** A storage cylinder for an ideal gas has a volume of $3.0 \times 10^{-4}\,m^3$. The gas is at a temperature of 23 °C and a pressure of $5.0 \times 10^7$ Pa.

**(i)** Show that the amount of gas in the cylinder is 6.1 mol.

[2]

**(ii)** The gas leaks slowly from the cylinder so that, after a time of 35 days, the pressure has reduced by 0.40%. The temperature remains constant.
Calculate the average rate, in atoms per second, at which gas atoms escape from the cylinder.

rate = ..................................................... $s^{-1}$ [4]

**18. O/N/14/43/Q3**

A fixed mass of gas has an initial volume of $5.00 \times 10^{-4}\,m^3$ at a pressure of $2.40 \times 10^5\,Pa$ and a temperature of 288 K. It is heated at constant pressure so that, in its final state, the volume is $14.5 \times 10^{-4}\,m^3$ at a temperature of 835 K, as illustrated in Fig. 3.1.

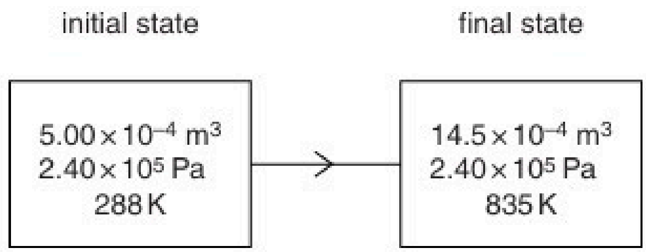


**Fig. 3.1**

**(a)** Show that these two states provide evidence that the gas behaves as an ideal gas.

[3]

**(b)** The total thermal energy supplied to the gas for this change is 569 J.

Determine

**(i)** the external work done,

work done = ........................................................ J [2]

**(ii)** the change in internal energy of the gas. State whether the change is an increase or a decrease in internal energy.

change in internal energy = ............................................................ J

............................................................................................................ [2]

**19. M/J/15/42/Q2**

In a sample of gas at room temperature, five atoms have the following speeds:

$$1.32 \times 10^3\,\mathrm{m\,s^{-1}}$$
$$1.50 \times 10^3\,\mathrm{m\,s^{-1}}$$
$$1.46 \times 10^3\,\mathrm{m\,s^{-1}}$$
$$1.28 \times 10^3\,\mathrm{m\,s^{-1}}$$
$$1.64 \times 10^3\,\mathrm{m\,s^{-1}}.$$

For these five atoms, calculate, to three significant figures,

**(a)** the mean speed,

mean speed = .................................................. $\mathrm{m\,s^{-1}}$ [1]

**(b)** the mean-square speed,

mean-square speed = .................................................. $\mathrm{m^2\,s^{-2}}$ [2]

**(c)** the root-mean-square speed.

root-mean-square speed = .................................................. $\mathrm{m\,s^{-1}}$ [1]

**20. O/N/15/41/Q2**

**(a)** State what is meant by an *ideal* gas.

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**(b)** The mean-square speed of the atoms of a fixed mass of an ideal gas at 32 °C is $1.9 \times 10^6\,m^2\,s^{-2}$.

The gas is heated at constant volume to a temperature of 80 °C.

Determine

**(i)** the rise, in kelvin, of the temperature of the gas,

temperature rise = ........................................................ K [1]

**(ii)** the root-mean-square (r.m.s.) speed of the atoms at 80 °C.

r.m.s. speed = ................................................ $m\,s^{-1}$ [3]

**21. O/N/15/43/Q2**

**(a)** An ideal gas is said to consist of molecules that are hard elastic identical spheres.

State two further assumptions of the kinetic theory of gases.

1. ..........................................................................................................................

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2. ..........................................................................................................................

.......................................................................................................................... [2]

**(b)** The number of molecules per unit volume in an ideal gas is $n$.

If it is assumed that all the molecules are moving with speed $v$, the pressure $p$ exerted by the gas on the walls of the vessel is given by

$$p = \frac{1}{3}nmv^2$$

where $m$ is the mass of one molecule.

Explain the reasoning by which this expression is modified to give the formula

$$p = \frac{1}{3}nm<c^2>.$$

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.......................................................................................................................... [1]

**(c)** The density of an ideal gas is $1.2\,\text{kg}\,\text{m}^{-3}$ at a pressure of $1.0 \times 10^5\,\text{Pa}$ and a temperature of 27 °C.

**(i)** Calculate the root-mean-square (r.m.s.) speed of the molecules of the gas at 27 °C.

r.m.s. speed = .................................................. $\text{m}\,\text{s}^{-1}$ [3]

**(ii)** Calculate the mean-square speed of the molecules at 207 °C.

mean-square speed = ............................................ $m^2 s^{-2}$ [2]

**22. F/M/16/42/Q2**

**(a)** State

**(i)** what is meant by *internal energy*,

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**(ii)** the basic assumption of the kinetic theory of gases that leads to the conclusion that there is zero potential energy between the molecules of an ideal gas.

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..........................................................................................................................................[1]

**(b)** The pressure $p$ and volume $V$ of an ideal gas are related by

$$pV = \frac{1}{3} Nm \langle c^2 \rangle$$

where $N$ is the number of molecules, $m$ is the mass of a molecule and $\langle c^2 \rangle$ is the mean-square speed of the molecules.

Use this equation to show that the mean kinetic energy $\langle E_K \rangle$ of a molecule is given by

$$\langle E_K \rangle = \frac{3}{2} kT$$

where $k$ is the Boltzmann constant and $T$ is the thermodynamic temperature.

[3]

**(c)** A cylinder contains 17 g of oxygen gas at a temperature of 12 °C. The mass of 1.0 mol of oxygen gas is 32 g. It may be assumed that the oxygen behaves as an ideal gas.

Calculate, for the oxygen gas in the cylinder,

**(i)** the mean kinetic energy of a molecule,

mean kinetic energy = .......................................................... J [2]

**(ii)** the number of molecules,

number = ............................................................ [2]

**(iii)** the total internal energy.

internal energy = .......................................................... J [1]

[Total: 11]

## 23. M/J/16/41/Q2

**(a)** An ideal gas is assumed to consist of atoms or molecules that behave as hard, identical spheres that are in continuous motion and undergo elastic collisions.

State two further assumptions of the kinetic theory of gases.

1. ..........................................................................................................................................

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2. ..........................................................................................................................................

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[2]

**(b)** Helium-4 ($^4_2$He) may be assumed to be an ideal gas.

**(i)** Show that the mass of one atom of helium-4 is $6.6 \times 10^{-24}$ g.

[1]

**(ii)** The mean kinetic energy $E_K$ of an atom of an ideal gas is given by the expression

$$E_K = \frac{3}{2}kT.$$

Calculate the root-mean-square (r.m.s.) speed of a helium-4 atom at a temperature of 27 °C.

r.m.s. speed = ................................................ m s$^{-1}$ [3]

[Total: 6]

**24. M/J/16/42/Q2**

**(a)** State what is meant by

**(i)** the Avogadro constant $N_A$,

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.......................................................................................................................................... [1]

**(ii)** the mole.

..........................................................................................................................................

.......................................................................................................................................... [2]

**(b)** A container has a volume of $1.8 \times 10^4\ \text{cm}^3$.

The ideal gas in the container has a pressure of $2.0 \times 10^7\ \text{Pa}$ at a temperature of 17 °C.

Show that the amount of gas in the cylinder is 150 mol.

[1]

**(c)** Gas molecules leak from the container in **(b)** at a constant rate of $1.5 \times 10^{19}\ \text{s}^{-1}$.
The temperature remains at 17 °C.
In a time $t$, the amount of gas in the container is found to be reduced by 5.0%.

Calculate

**(i)** the pressure of the gas after the time $t$,

pressure = ..................................................... Pa [2]

**(ii)** the time $t$.

$t$ = ............................................................ s [3]

[Total: 9]

## 25. O/N/16/42/Q2

**(a)** The equation of state for an ideal gas of volume $V$ at pressure $p$ is

$$pV = nRT$$

where $R$ is the molar gas constant.

State what is meant by

**(i)** the symbol $n$,

..........................................................................................................................................

..........................................................................................................................................[1]

**(ii)** the symbol $T$.

..........................................................................................................................................

..........................................................................................................................................[1]

**(b)** An ideal gas is held in a container of volume $2.4 \times 10^3\,\text{cm}^3$ at pressure $4.9 \times 10^5\,\text{Pa}$.
The temperature of the gas is 100 °C.

Show that the number of molecules of the gas in the container is $2.3 \times 10^{23}$.

[3]

**(c)** Use data from **(b)** to estimate the mean distance between molecules in the gas.

mean distance = ..................................................... cm [3]

**26. M/J/17/41/Q4**

**(a)** Describe the motion of molecules in a gas, according to the kinetic theory of gases.

..........................................................................................................................................

..........................................................................................................................................

.....................................................................................................................................[2]

**(b)** Describe what is observed when viewing Brownian motion that provides evidence for your answer in **(a)**.

..........................................................................................................................................

..........................................................................................................................................

.....................................................................................................................................[2]

**(c)** At a pressure of $1.05 \times 10^5$ Pa and a temperature of 27 °C, 1.00 mol of helium gas has a volume of $0.0240\,m^3$.
The mass of 1.00 mol of helium gas, assumed to be an ideal gas, is 4.00 g.

**(i)** Calculate the root-mean-square (r.m.s.) speed of an atom of helium gas for a temperature of 27 °C.

r.m.s. speed = .................................................. $m\,s^{-1}$ [3]

**(ii)** Using your answer in **(i)**, calculate the r.m.s. speed of the atoms at 177 °C.

r.m.s. speed = .................................................. $m\,s^{-1}$ [3]

**27. M/J/17/42/Q2**

**(a)** The pressure $p$ and volume $V$ of an ideal gas are related to the density $\rho$ of the gas by the expression

$$p = \frac{1}{3}\rho\langle c^2 \rangle.$$

**(i)** State what is meant by the symbol $\langle c^2 \rangle$.

..........................................................................................................................................

..........................................................................................................................................[1]

**(ii)** Use the expression to show that the mean kinetic energy $E_K$ of a gas molecule is given by

$$E_K = \frac{3}{2}kT$$

where $k$ is the Boltzmann constant and $T$ is the thermodynamic temperature.

[3]

**(b) (i)** An ideal gas containing 1.0 mol of molecules is heated at constant volume.
Use the expression in **(a)(ii)** to show that the thermal energy required to raise the temperature of the gas by 1.0 K has a value of $\frac{3}{2}R$, where $R$ is the molar gas constant.

[3]

**(ii)** Nitrogen may be assumed to be an ideal gas. The molar mass of nitrogen gas is $28\,g\,mol^{-1}$.
Use the answer in **(b)(i)** to calculate a value for the specific heat capacity, in $J\,kg^{-1}\,K^{-1}$, at constant volume for nitrogen.

specific heat capacity = .......................................... $J\,kg^{-1}\,K^{-1}$ [2]

[Total: 9]

**28. F/M/18/42/Q2**

A cylinder contains 5.12 mol of an ideal gas at pressure of $5.60 \times 10^5$ Pa and volume $3.80 \times 10^4$ cm$^3$.

**(a)** Determine the temperature of the gas.

temperature = .......................................................... K [2]

**(b)** The average kinetic energy $E_K$ of a molecule of the gas is given by the expression

$$E_K = \frac{3}{2}kT$$

where $k$ is the Boltzmann constant and $T$ is the thermodynamic temperature.

The gas is heated at constant pressure so that its temperature rises by 125 K.

**(i)** Use your answer in **(a)** to determine the new volume of the gas.

volume = ...................................................... cm$^3$ [2]

**(ii)** Calculate the increase in internal energy of the gas. Explain your working.

increase in internal energy = .......................................................... J [3]

**(c)** **(i)** Use your answer in **(b)(i)** to determine the external work done during the expansion of the gas.

work done = ........................................................ J [2]

**(ii)** Calculate the total thermal energy required to heat the gas in **(b)**.

energy = ........................................................ J [1]

[Total: 10]

**29. M/J/18/42/Q2**

**(a)** Use one of the assumptions of the kinetic theory of gases to explain why the potential energy of the molecules of an ideal gas is zero.

..........................................................................................................................................

.....................................................................................................................................[1]

**(b)** The average translational kinetic energy $E_K$ of a molecule of an ideal gas is given by the expression

$$E_K = \frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$$

where $m$ is the mass of a molecule and $k$ is the Boltzmann constant.

State the meaning of the symbol

**(i)** $\langle c^2 \rangle$,

.....................................................................................................................................[1]

**(ii)** $T$.

.....................................................................................................................................[1]

**(c)** A cylinder of constant volume $4.7 \times 10^4\,cm^3$ contains an ideal gas at pressure $2.6 \times 10^5\,Pa$ and temperature 173 °C.

The gas is heated. The thermal energy transferred to the gas is 2900 J. The final temperature and pressure of the gas are $T$ and $p$, as illustrated in Fig. 2.1.


**Fig. 2.1**

**(i)** Calculate

**1.** the number $N$ of molecules in the cylinder,

$N$ = ...........................................................[3]

**2.** the increase in average kinetic energy of a molecule during the heating process.

increase = ........................................................ J [1]

**(ii)** Use your answer in **(i) part 2** to determine the final temperature $T$, in kelvin, of the gas in the cylinder.

$T$ = ........................................................ K [3]

[Total: 10]

***30. O/N/18/42/Q2***

**(a)** State what is meant by an *ideal gas*.

.............................................................................................................................................

.............................................................................................................................................

.......................................................................................................................................[2]

**(b)** An ideal gas comprised of single atoms is contained in a cylinder and has a volume of $1.84 \times 10^{-2}\,\text{m}^3$ at a pressure of $2.12 \times 10^7\,\text{Pa}$.
The mass of gas in the cylinder is 3.20 kg.

**(i)** Determine, to three significant figures, the root-mean-square (r.m.s.) speed of the atoms of the gas.

r.m.s. speed = ................................................... $\text{m s}^{-1}$ [3]

**(ii)** The temperature of the gas in the cylinder is 22 °C.

Determine, to three significant figures,

**1.** the amount, in mol, of the gas,

amount = .................................................... mol [2]

**2.** the mass of one atom of the gas.

mass = ...................................................... kg [2]

**(c)** Use your answer in **(b)(ii) part 2** to determine the nucleon number *A* of an atom of the gas.

*A* = ........................................................... [1]

[Total: 10]

**31. F/M/19/42/Q2**

The pressure $p$ of an ideal gas having density $\rho$ is given by the expression

$$p = \frac{1}{3}\rho\langle c^2\rangle.$$

**(a)** State what is meant by:

**(i)** an ideal gas

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(ii)** the symbol $\langle c^2\rangle$.

..........................................................................................................................................

.................................................................................................................................... [1]

**(b)** A cylinder contains a fixed mass of a gas at a temperature of 120 °C. The gas has a volume of $6.8 \times 10^{-3}\,m^3$ at a pressure $2.4 \times 10^5\,Pa$.

**(i)** Assuming the gas acts like an ideal gas, show that the number of atoms of gas in the cylinder is $3.0 \times 10^{23}$.

[3]

**(ii)** Each atom of the gas, assumed to be a sphere, has a radius of $3.2 \times 10^{-11}\,m$.

Use the answer in **(i)** to estimate the actual volume occupied by the gas atoms.

volume = ....................................................... $m^3$ [2]

**(iii)** One of the assumptions of the kinetic theory of gases is related to the volume of the atoms.
State this assumption. Explain whether your answer in **(ii)** is consistent with this assumption.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

[Total: 10]

**32. M/J/19/41/Q2**

A fixed mass of an ideal gas has volume $210\,cm^3$ at pressure $3.0 \times 10^5\,Pa$ and temperature 270 K.

The volume of the gas is reduced at constant pressure to $140\,cm^3$, as shown in Fig. 2.1.


**Fig. 2.1**

The final temperature of the gas is $T$.

**(a)** Determine:

**(i)** the amount of gas

amount = .................................................... mol [3]

**(ii)** the final temperature $T$ of the gas

$T$ = ........................................................K [2]

**(iii)** the external work done on the gas.

work done = ....................................................... J [2]

**(b)** For this change in volume and temperature of the gas, the thermal energy transferred is 53 J.

Determine $\Delta U$, the change in internal energy of the gas.

$\Delta U$ = .......................................................... J [3]

[Total: 10]

***33. O/N/19/41/Q2***

**(a)** The kinetic theory of gases is based on a number of assumptions about the molecules of a gas.

State the assumption that is related to the volume of the molecules of the gas.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(b)** An ideal gas occupies a volume of $2.40 \times 10^{-2}\,m^3$ at a pressure of $4.60 \times 10^5\,Pa$ and a temperature of 23 °C.

**(i)** Calculate the number of molecules in the gas.

number = .......................................................... [3]

**(ii)** Each molecule has a diameter of approximately $3 \times 10^{-10}\,m$.

Estimate the total volume of the gas molecules.

volume = ...................................................... $m^3$ [3]

**(c)** By reference to your answer in **(b)(ii)**, suggest why the assumption in **(a)** is justified.

..........................................................................................................................................

.................................................................................................................................... [1]

**34. O/N/19/42/Q2**

**(a)** Smoke particles are suspended in still air. Brownian motion of the smoke particles is seen through a microscope.

Describe:

**(i)** what is seen through the microscope

..........................................................................................................................................

.................................................................................................................................... [1]

**(ii)** how Brownian motion provides evidence for the nature of the movement of gas molecules.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(b)** A fixed mass of an ideal gas has volume $2.40 \times 10^3\,cm^3$ at pressure $3.51 \times 10^5\,Pa$ and temperature 290 K. The gas is heated at constant volume until the temperature is 310 K at pressure $3.75 \times 10^5\,Pa$, as illustrated in Fig. 2.1.


**Fig. 2.1**

The quantity of thermal energy required to raise the temperature of 1.00 mol of the gas by 1.00 K at constant volume is 12.5 J.

Calculate, to three significant figures:

**(i)** the amount, in mol, of the gas

amount = .................................................... mol [3]

**(ii)** the thermal energy transfer during the change.

energy transfer = ........................................................ J [2]

**(c)** For the change in the gas in **(b)**, state:

**(i)** the quantity of external work done on the gas

work done = ........................................................ J [1]

**(ii)** the change in internal energy, with the direction of this change.

change = ............................................................. J

direction ..................................................................

[2]

[Total: 11]

**35. F/M/20/42/Q2**

A large container of volume $85\,m^3$ is filled with 110 kg of an ideal gas. The pressure of the gas is $1.0 \times 10^5$ Pa at temperature $T$.

The mass of 1.0 mol of the gas is 32 g.

**(a)** Show that the temperature $T$ of the gas is approximately 300 K.

[3]

**(b)** The temperature of the gas is increased to 350 K at constant volume. The specific heat capacity of the gas for this change is $0.66\,J\,kg^{-1}\,K^{-1}$.

Calculate the energy supplied to the gas by heating.

energy = ........................................................ J [2]

**(c)** Explain how movement of the gas molecules causes pressure in the container.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [3]

**(d)** The temperature of a gas depends on the root-mean-square (r.m.s.) speed of its molecules.

Calculate the ratio:

$$\frac{\text{r.m.s. speed of gas molecules at } 350\,\text{K}}{\text{r.m.s. speed of gas molecules at } 300\,\text{K}}.$$

ratio = ........................................................ [2]

[Total: 10]

**36. M/J/20/42/Q2**

**(a)** A square box of volume $V$ contains $N$ molecules of an ideal gas. Each molecule has mass $m$.

Using the kinetic theory of ideal gases, it can be shown that, if all the molecules are moving with speed $v$ at right angles to one face of the box, the pressure $p$ exerted on the face of the box is given by the expression

$$pV = Nmv^2. \qquad \text{(equation 1)}$$

This expression leads to the formula

$$p = \frac{1}{3}\rho\langle c^2\rangle \qquad \text{(equation 2)}$$

for the pressure $p$ of an ideal gas, where $\rho$ is the density of the gas and $\langle c^2\rangle$ is the mean-square speed of the molecules.

Explain how each of the following terms in equation 2 is derived from equation 1:

$\rho$: ..........................................................................................................................................

..........................................................................................................................................

$\frac{1}{3}$: ..........................................................................................................................................

..........................................................................................................................................

$\langle c^2\rangle$: ..........................................................................................................................................

..........................................................................................................................................
[4]

**(b)** An ideal gas has volume, pressure and temperature as shown in Fig. 2.1.

| |
|---|
| volume $6.0 \times 10^{-3}\,m^3$ |
| pressure $3.0 \times 10^5\,Pa$ |
| temperature $17\,°C$ |

**Fig. 2.1**

The mass of the gas is 20.7 g.

Calculate the mass of one molecule of the gas.

mass = ...................................................... g [4]

**37. O/N/20/42/Q2**

**(a)** State what is meant by the *internal energy* of a system.

..................................................................................................................................................

..................................................................................................................................................

.............................................................................................................................................. [2]

**(b)** The atoms of an ideal gas occupy a container of volume $2.30 \times 10^{-3}\,m^3$ at pressure $2.60 \times 10^5\,Pa$ and temperature 180 K, as illustrated in Fig. 2.1.

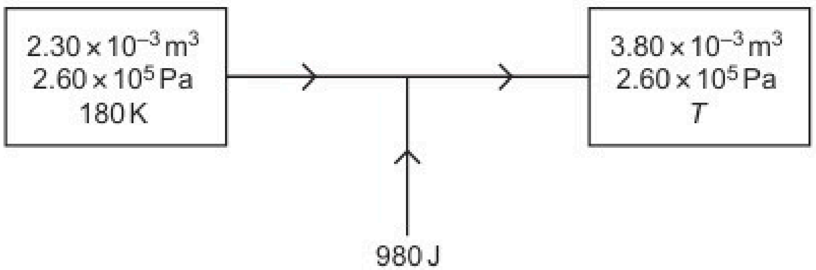


**Fig. 2.1**

The gas is heated at constant pressure so that its volume becomes $3.80 \times 10^{-3}\,m^3$ at a temperature $T$.

For the fixed mass of gas, calculate:

**(i)** the amount of substance, in mol

amount = ...................................................... mol [2]

**(ii)** the temperature $T$, in K.

$T$ = ...................................................... K [2]

**(c)** During the change in **(b)**, the thermal energy supplied to the gas is 980 J.

**(i)** Determine the work done on the gas during this change. Explain your working.

work done = ........................................................ J [3]

**(ii)** Determine the change $\Delta U$ in internal energy of the gas.

$\Delta U$ = ........................................................ J [1]

[Total: 10]

### 38. F/M/21/42/Q2

A fixed mass of an ideal gas is at a temperature of 21 °C. The pressure of the gas is $2.3 \times 10^5$ Pa and its volume is $3.5 \times 10^{-3}$ m$^3$.

**(a)** **(i)** Calculate the number $N$ of molecules in the gas.

$N$ = ............................................................ [2]

**(ii)** The mass of one molecule of the gas is 40 u.

Determine the root-mean-square (r.m.s.) speed of the gas molecules.

r.m.s. speed = .................................................. m s$^{-1}$ [2]

**(b)** The temperature of the gas is increased by 84 °C.

Calculate the value of the ratio

$$\frac{\text{new r.m.s. speed of molecules}}{\text{original r.m.s. speed of molecules}}.$$

ratio = ............................................................ [2]

[Total: 6]

### 39. M/J/21/41/Q2

An ideal gas is contained in a cylinder by means of a movable frictionless piston, as illustrated in Fig. 2.1.

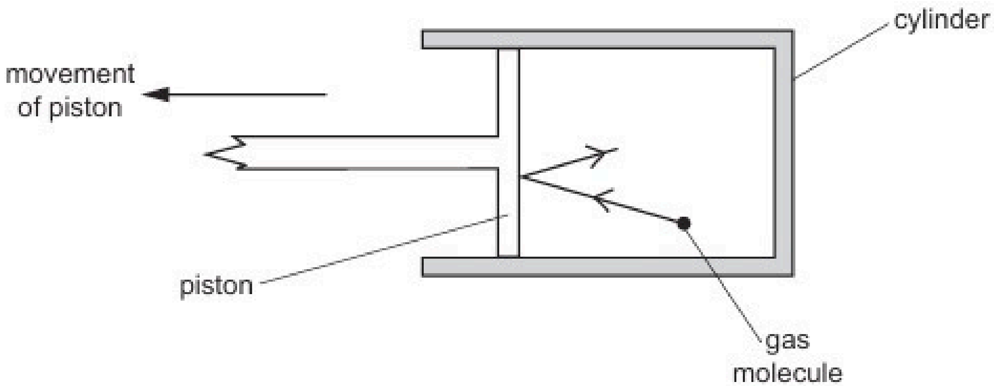


**Fig. 2.1**

Initially, the gas has a volume of $1.8 \times 10^{-3}\,m^3$ at a pressure of $3.3 \times 10^5\,Pa$ and a temperature of 310K.

**(a)** Show that the number of gas molecules in the cylinder is $1.4 \times 10^{23}$.

[2]

**(b)** Use kinetic theory to explain why, when the piston is moved so that the gas expands, this causes a decrease in the temperature of the gas.

...................................................................................................................................

...................................................................................................................................

...................................................................................................................................

............................................................................................................................... [3]

**(c)** The gas expands so that its volume increases to $2.4 \times 10^{-3}\,m^3$ at a pressure of $2.3 \times 10^5\,Pa$ and a temperature of 288 K, as shown in Fig. 2.2.

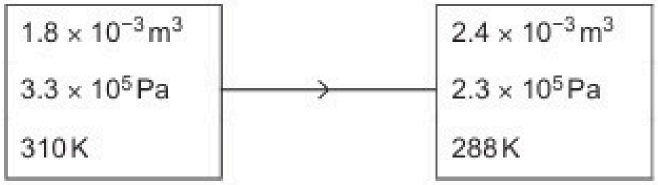


**Fig. 2.2**

**(i)** The average translational kinetic energy $E_K$ of a molecule of an ideal gas is given by

$$E_K = \frac{3}{2}kT$$

where $k$ is the Boltzmann constant and $T$ is the thermodynamic temperature.

Calculate the increase in internal energy $\Delta U$ of the gas during the expansion.

$\Delta U$ = ........................................................ J [3]

**(ii)** The work done by the gas during the expansion is 76 J.

Use your answer in **(i)** to explain whether thermal energy is transferred to or from the gas during the expansion.

...........................................................................................................................................

...........................................................................................................................................

..................................................................................................................................... [2]

[Total: 10]

## 40. M/J/21/42/Q2

An ideal gas has a volume of $3.1 \times 10^{-3}\,m^3$ at a pressure of $8.5 \times 10^5\,Pa$ and a temperature of 290 K, as shown in Fig. 2.1.

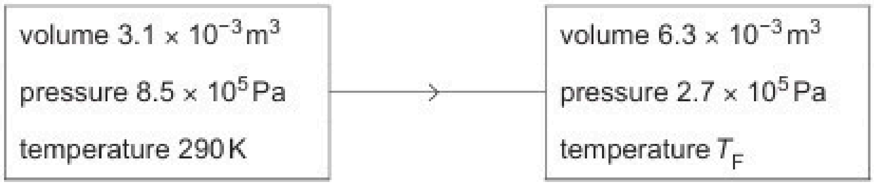


**Fig. 2.1**

The gas suddenly expands to a volume of $6.3 \times 10^{-3}\,m^3$. During the expansion, no thermal energy is transferred. The final pressure of the gas is $2.7 \times 10^5\,Pa$ at temperature $T_F$, as shown in Fig. 2.1.

**(a)** Show that the number of gas molecules is $6.6 \times 10^{23}$.

[3]

**(b)** **(i)** Show that the final temperature $T_F$ of the gas is 190 K.

[1]

**(ii)** The average translational kinetic energy $E_K$ of a molecule of an ideal gas is given by

$$E_K = \frac{3}{2}kT$$

where $T$ is the thermodynamic temperature and $k$ is the Boltzmann constant.

Calculate the increase in internal energy $\Delta U$ of the gas.

$\Delta U$ = ........................................................ J [3]

**(c)** Use the first law of thermodynamics to explain why the external work $w$ done on the gas during the expansion is equal to the increase in internal energy in **(b)(ii)**.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

[Total: 9]

**41. O/N/21/42/Q3**

**(a)** One of the assumptions of the kinetic theory of gases is that all collisions involving molecules of the gas are elastic.

**(i)** State what is meant by an *elastic* collision.

..........................................................................................................................................

.......................................................................................................................................... [1]

**(ii)** State **two** other assumptions of the kinetic theory of gases.

1. ..........................................................................................................................................

..........................................................................................................................................

2. ..........................................................................................................................................

..........................................................................................................................................

[2]

**(b)** A molecule of an ideal gas has mass $m$ and is contained in a cubic box of side length $L$. The molecule is moving with velocity $u$ towards the face of the box that is shaded in Fig. 3.1.

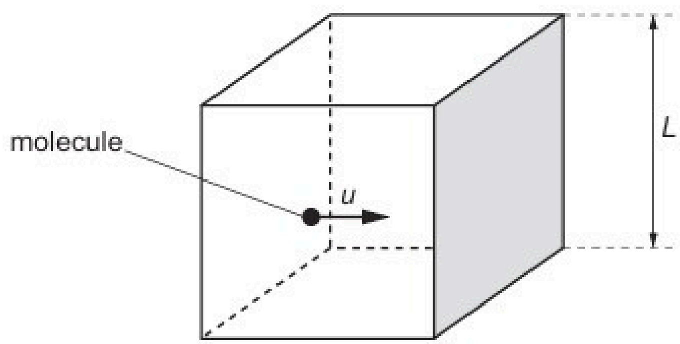


**Fig. 3.1**

The molecule collides elastically with the shaded face and the face opposite to it alternately.

Deduce expressions, in terms of $m$, $u$ and $L$, for:

**(i)** the magnitude of the change in momentum of the molecule on colliding with a face

change in momentum = .......................................................... [1]

**(ii)** the time between consecutive collisions of the molecule with the shaded face

time = .......................................................... [1]

**(iii)** the average force exerted by the molecule on the shaded face

force = .......................................................... [1]

**(iv)** the pressure on the shaded face if the force in **(iii)** is exerted over the whole area of the face.

pressure = .......................................................... [1]

**(c)** When the model described in **(b)** is extended to three dimensions, and to a gas containing $N$ molecules, each of mass $m$, travelling with mean-square speed $\langle c^2 \rangle$, it can be shown that

$$pV = \tfrac{1}{3}Nm\langle c^2 \rangle$$

where $p$ is the pressure exerted by the gas and $V$ is the volume of the gas.

Use this expression, together with the equation of state of an ideal gas, to show that the average translational kinetic energy $E_K$ of a molecule of an ideal gas is given by

$$E_K = \tfrac{3}{2}kT$$

where $T$ is the thermodynamic temperature of the gas and $k$ is the Boltzmann constant.

[2]

**(d)** The mass of a hydrogen molecule is $3.34 \times 10^{-27}$ kg.

Use the expression for $E_K$ in **(c)** to determine the root-mean-square (r.m.s.) speed of a molecule of hydrogen gas at 25 °C.

r.m.s. speed = ................................................ $m\,s^{-1}$ [2]

## *42. O/N/22/42/Q3*

**(a)** The equation of state for an ideal gas can be written as

$$pV = NkT.$$

State the meaning of each of the symbols in this equation.

$p$: ..............................................................................................................................

$V$: ..............................................................................................................................

$N$: ..............................................................................................................................

$k$: ..............................................................................................................................

$T$: ..............................................................................................................................

[3]

**(b)** Use the equation in **(a)** to show that the average translational kinetic energy $E_K$ of a molecule of an ideal gas is given by

$$E_K = \frac{3}{2}kT.$$

[2]

**(c)** The mass of an oxygen molecule is $5.31 \times 10^{-26}$ kg. Assume that oxygen behaves as an ideal gas.

**(i)** Use the equation in **(b)** to determine the root-mean-square (r.m.s.) speed $u$ of an oxygen molecule at 23 °C.

$u$ = ................................................ $m s^{-1}$ [3]

**(ii)** A fixed mass of oxygen gas at initial pressure $P$ is sealed in a cylindrical container by a movable piston at one end, as shown in Fig. 3.1.

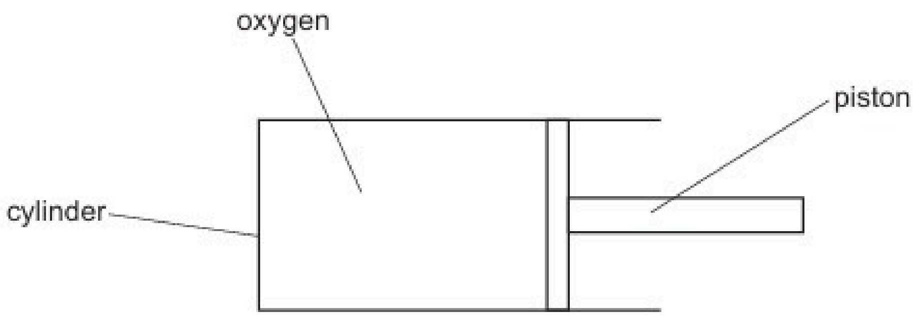


**Fig. 3.1**

The temperature of the gas is 23 °C.

The piston is slowly moved into the cylinder so that the oxygen gas is compressed. At all times, the gas and the container remain in thermal equilibrium with the surroundings.

On Fig. 3.2, sketch the variation with pressure of the r.m.s. speed of the oxygen molecules as the pressure increases.


**Fig. 3.2**

[2]

[Total: 10]

### 43. M/J/23/42/Q2

(a) (i) State what is meant by an ideal gas.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

(ii) State the temperature, in degrees Celsius, of absolute zero.

temperature = ...................................................... °C [1]

(b) A sealed vessel contains a mass of 0.0424 kg of an ideal gas at 227 °C.
The pressure of the gas is $1.37 \times 10^5$ Pa and the volume of the gas is $0.640\,m^3$.

Calculate:

(i) the number of molecules of the gas in the vessel

number of molecules = ........................................................... [3]

(ii) the mass of one molecule of the gas

mass = ...................................................... kg [1]

(iii) the root-mean-square (r.m.s.) speed $v$ of the molecules of the gas.

$v$ = ................................................ $m\,s^{-1}$ [3]

**(c)** The gas in **(b)** is now cooled gradually to absolute zero.

On Fig. 2.1, sketch the variation with thermodynamic temperature $T$ of the r.m.s. speed of the molecules of the gas.

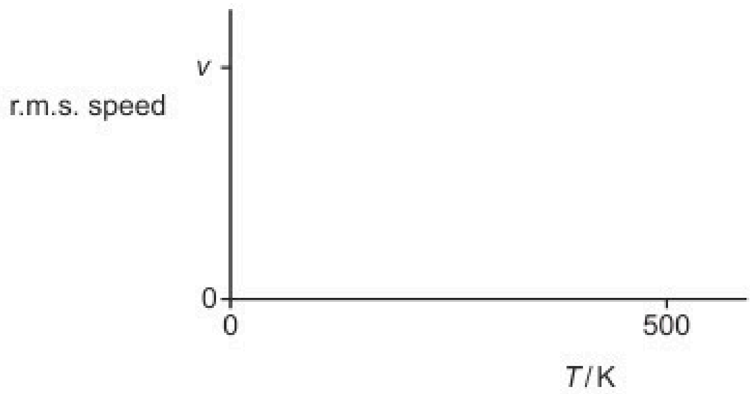


**Fig. 2.1**

[2]

[Total: 12]

# *ANSWERS*

***1.***

**(a)** **(i)** $pV = nRT$

$V = (8.31 \times 300)/(1.02 \times 10^5)$ ........ C1

$= 0.0244\ m^3$ (if uses Celsius, then 0/2) ........ A1

**(ii)** volume occupied by one atom $= 0.0244 / (6.02 \times 10^{23}) = 4.06 \times 10^{-26}\ m^3$ ........ M1

separation $\approx \sqrt[3]{(4.06 \times 10^{-26})}$ ........ A1

$= 3.44 \times 10^{-9}\ m$ ........ A0

**(b)** **(i)** $F = GMm / r^2$ ........ C1

$= (6.67 \times 10^{-11} \times \{4 \times 1.66 \times 10^{-27}\}^2) / (3.44 \times 10^{-9})^2$ ........ C1

$= 2.49 \times 10^{-46}\ N$ ........ A1

**(ii)** ratio $= (4 \times 1.66 \times 10^{-27} \times 9.8) / 2.49 \times 10^{-46}$ ........ C1

$= 2.6 \times 10^{20}$ ........ A1

**(c)** assumption that forces between atoms are negligible ........ B1

comment e.g. ratio shows gravitational force to be very small

e.g. force is very much less than weight

e.g. if there are forces, they are not gravitational ........ B1

***2.***

**(a)** molecule(s) rebound from wall of vessel / hits walls B1

change in momentum gives rise to impulse / force B1

*either* (many impulses) averaged to give constant force / pressure

*or* the molecules are in random motion B1

**(b)** (i) $p = \frac{1}{3}\rho\langle c^2\rangle$ C1

$1.02 \times 10^5 = \frac{1}{3} \times 0.900 \times \langle c^2\rangle$

$\langle c^2\rangle = 3.4 \times 10^5$ C1

$c_{RMS} = 580\ m\ s^{-1}$ A1

(ii) *either* $\langle c^2\rangle \propto T$ *or* $\langle c^2\rangle = 2 \times 3.4 \times 10^5$ C1

$c_{RMS} = 830\ m\ s^{-1}$ (allow 820) A1

**(c)** $c_{RMS}$ depends on temperature (alone) B1

so no effect B1

***3.***

**(a)** *either* $pV = NkT$ *or* $pV = nRT$ <u>*and*</u> $n = N / N_A$ .................................................. C1
clear correct substitution e.g.
$2.5 \times 10^5 \times 4.5 \times 10^3 \times 10^{-6} = N \times 1.38 \times 10^{-23} \times 290$ .............................................M1
$N = 2.8 \times 10^{23}$ ........................................................................................................ A0
(*allow 1 mark for calculation of* $n = 0.467$ *mol*)

**(b)** **(i)** volume $= (1.2 \times 10^{-10})^3 \times 2.8 \times 10^{23}$ *or* $\frac{4}{3}\pi r^3 \times 2.8 \times 10^{23}$ .............................. C1
$= 4.8 \times 10^{-7}\ \text{m}^3$ $\quad$ $2.53 \times 10^{-7}\ \text{m}^3$ ...................................... A1

**(ii)** *either* $4.5 \times 10^3\ \text{cm}^3 >> 0.48\ \text{cm}^3$ *or* ratio of volumes is about $10^{-4}$ ................ B1
justified because volume of molecules is negligible ............................................ B1

***4.***

**(a)** **(i)** 27.2 + 273.15 or 27.2 + 273.2 C1
300.4 K A1

**(ii)** 11.6 K A1

**(b)** **(i)** ($<c^2>$ is the) mean / average square speed B1

**(ii)** $\rho = Nm/V$ with <u>$N$ explained</u> B1
so, $pV = 1/3\ Nm<c^2>$ B1
and $pV = NkT$ with <u>$k$ explained</u> B1
<u>so mean kinetic energy / $<E_K> = \frac{1}{2}m<c^2> = 3/2\ kT$</u> B1

**(c)** **(i)** $pV = nRT$
$2.1 \times 10^7 \times 7.8 \times 10^{-3} = n \times 8.3 \times 290$ C1
$n = 68$ mol A1

**(ii)** mean kinetic energy $= 3/2\ kT$
$= 3/2 \times 1.38 \times 10^{-23} \times 290$ C1
$= 6.0 \times 10^{-21}$ J A1

**(iii)** realisation that total internal energy is the total kinetic energy C1
energy $= 6.0 \times 10^{-21} \times 68 \times 6.02 \times 10^{23}$ C1
$= 2.46 \times 10^5$ J A1

*5.*

**(a)** atoms / molecules / particles behave as elastic (identical) spheres (1)
volume of atoms / molecules negligible compared to volume of containing vessel (1)
time of collision negligible to time between collisions (1)
no forces of attraction or repulsion between atoms / molecules (1)
atoms / molecules / particles are in (continuous) random motion (1)
(*any four, 1 each*) B4

**(b)** $pV = \frac{1}{3}Nm\langle c^2\rangle$ and $pV = nRT$ *or* $pV = NkT$ B1

$\frac{1}{3}Nm\langle c^2\rangle = nRT$ or $= NkT$ and $\langle E_K\rangle = ½m\langle c^2\rangle$ B1

$n = N/N_A$ *or* $k = R/N_A$ B1

$\langle E_K\rangle = \frac{3}{2} \times R/N_A \times T$ A0

**(c) (i)** reaction represents *either* build-up of nucleus from light nuclei
*or* build-up of heavy nucleus from nuclei M1
so fusion reaction A1

**(ii)** proton and deuterium nucleus will have equal kinetic energies B1
$1.2 \times 10^{-14} = \frac{3}{2} \times 8.31 / (6.02 \times 10^{23}) \times T$ C1
$T = 5.8 \times 10^8$ K A1
(*use of* $E = 2.4 \times 10^{-14}$ *giving* $1.16 \times 10^9$ *K scores 1 mark*)

**(iii)** *either* inter-molecular / atomic / nuclear forces exist
*or* proton and deuterium nucleus are positively charged / repel B1

*6.*

**(a)** number of atoms of carbon-12 M1
in 0.012 kg of carbon-12 A1

**(b)** $pV = NkT$ or $pV = nRT$ C1
substitutes temperature as 298 K C1
*either* $1.1 \times 10^5 \times 6.5 \times 10^{-2} = N \times 1.38 \times 10^{-23} \times 298$
*or* $1.1 \times 10^5 \times 6.5 \times 10^{-2} = n \times 8.31 \times 298$ and $n = N / 6.02 \times 10^{23}$ C1
$N = 1.7 \times 10^{24}$ A1

*7.*

**(a)** amount of substance — M1
containing same number of particles as in 0.012 kg of carbon-12 — A1

**(b)** $pV = nRT$ — C1
amount = $(2.3 \times 10^5 \times 3.1 \times 10^{-3}) / (8.31 \times 290)$
$+ (2.3 \times 10^5 \times 4.6 \times 10^{-3}) / (8.31 \times 303)$ — C1
$= 0.296 + 0.420$ — C1
$= 0.716$ mol — A1
*(give full credit for starting equation $pV = NkT$ and $N = nN_A$)*

*8.*

**(a)** e.g. moving in random (rapid) motion of <u>molecules/atoms/particles</u>
no intermolecular forces of attraction/repulsion
volume of <u>molecules/atoms/particles</u> negligible <u>compared</u> to volume of container
time of collision negligible to time between collisions
*(1 each, max 2)* — B2

**(b) (i) 1.** number of (gas) <u>molecules</u> — B1

**2.** mean square speed/velocity (of gas molecules) — B1

**(ii)** either $pV = NkT$ *or* $pV = nRT$ and links $n$ and $k$
and $\langle E_K \rangle = \frac{1}{2}m\langle c^2 \rangle$ — M1
clear algebra leading to $\langle E_K \rangle = \frac{3}{2}kT$ — A1

**(c) (i)** sum of potential energy and kinetic energy of <u>molecules/atoms/particles</u> — M1
reference to random (distribution) — A1

**(ii)** no intermolecular forces so no potential energy — B1
(change in) internal energy is (change in) kinetic energy and this is proportional to (change in ) $T$ — B1

***9.***

**(a) (i)** weight = $GMm/r^2$ C1

$= (6.67 \times 10^{-11} \times 6.42 \times 10^{23} \times 1.40)/(½ \times 6.79 \times 10^{6})^2$ C1

$= 5.20$ N A1

**(ii)** potential energy = $-GMm/r$ C1

$= -(6.67 \times 10^{-11} \times 6.42 \times 10^{23} \times 1.40)/(½ \times 6.79 \times 10^{6})$ M1

$= -1.77 \times 10^{7}$ J A0

**(b)** *either* $½mv^2 = 1.77 \times 10^7$ C1

$v^2 = (1.77 \times 10^7 \times 2)/1.40$ C1

$v = 5.03 \times 10^3$ m s$^{-1}$ A1

*or* $½mv^2 = GMm/r$ (C1)

$v^2 = (2 \times 6.67 \times 10^{-11} \times 6.42 \times 10^{23})/(6.79 \times 10^6/2)$ (C1)

$v = 5.02 \times 10^3$ m s$^{-1}$ (A1)

**(c) (i)** $½ \times 2 \times 1.66 \times 10^{-27} \times (5.03 \times 10^3)^2 = \frac{3}{2} \times 1.38 \times 10^{-23} \times T$ C1

$T = 2030$ K A1

**(ii)** *either* because there is a range of speeds M1

some molecules have a higher speed A1

*or* some escape from point above planet surface (M1)

so initial potential energy is higher (A1)

***10.***

**(a) (i)** *either* random motion

*or* constant velocity until hits wall/other molecule B1

**(ii)** (total) volume of molecules is negligible M1

compared to volume of containing vessel A1

*or*

radius/diameter of a molecule is negligible (M

compared to the average intermolecular distance (A

**(b)** *either* molecule has component of velocity in three directions

*or* $c^2 = c_X^2 + c_Y^2 + c_Z^2$ M1

random motion and averaging, so $<c_X^2> = <c_Y^2> = <c_Z^2>$ M1

$<c^2> = 3<c_X^2>$ A1

so, $pV = ⅓Nm<c^2>$ A0

**(c)** $<c^2> \propto T$ or $c_{rms} \propto \sqrt{T}$ C1

temperatures are 300 K and 373 K C1

$c_{rms} = 580$ m s$^{-1}$ A1

***11.***

**(a)** **(i)** number of molecules B1

**(ii)** mean square speed B1

**(b)** **(i)** **1.** $pV = nRT$ C1
$n = (6.1 \times 10^5 \times 2.1 \times 10^4 \times 10^{-6}) / (8.31 \times 285)$ C1
$n = 5.4\,\text{mol}$ A1

**2.** *either* $N = nN_A$
$= 5.4 \times 6.02 \times 10^{23}$ C1
$= 3.26 \times 10^{24}$ A1
*or*
$pV = NkT$
$N = (6.1 \times 10^5 \times 2.1 \times 10^4 \times 10^{-6}) / (1.38 \times 10^{-23} \times 285)$ (C1)
$N = 3.26 \times 10^{24}$ (A1)

**(ii)** *either* $6.1 \times 10^5 \times 2.1 \times 10^{-2} = {}^1/_3 \times 3.25 \times 10^{24} \times 4 \times 1.66 \times 10^{-27} \times \langle c^2 \rangle$ C1
$\langle c^2 \rangle = 1.78 \times 10^6$ C1
$c_{RMS} = 1.33 \times 10^3\,\text{m s}^{-1}$ A1
*or*
${}^1/_2 \times 4 \times 1.66 \times 10^{-27} \times \langle c^2 \rangle = {}^3/_2 \times 1.38 \times 10^{-23} \times 285$ (C1)
$\langle c^2 \rangle = 1.78 \times 10^6$ (C1)
$c_{RMS} = 1.33 \times 10^3\,\text{m s}^{-1}$ (A1)

***12.***

**(a)** obeys the equation $pV$ = constant $\times$ $T$ or $pV = nRT$ M1
$p$, $V$ and $T$ explained A1
at all values of $p$, $V$ and $T$/fixed mass/$n$ is constant A1

**(b)** **(i)** $3.4 \times 10^5 \times 2.5 \times 10^3 \times 10^{-6} = n \times 8.31 \times 300$ M1
$n = 0.34\,\text{mol}$ A0

**(ii)** for total mass/amount of gas
$3.9 \times 10^5 \times (2.5 + 1.6) \times 10^3 \times 10^{-6} = (0.34 + 0.20) \times 8.31 \times T$ C1
$T = 360\,\text{K}$ A1

**(c)** when tap opened
gas passed (from cylinder B) to cylinder A B1
work done <u>on</u> gas in cylinder A (and no heating) M1
so internal energy and hence temperature increase A1

*13.*

**(a)** **(i)** **1.** $pV = nRT$

$1.80 \times 10^{-3} \times 2.60 \times 105 = n \times 8.31 \times 297$ C1

$n = 0.19$ mol A1

**2.** $\Delta q = mc\Delta T$

$95.0 = 0.190 \times 12.5 \times \Delta T$ B1

$\Delta T = 40$ K A1

*(allow 2 marks for correct answer with clear logic shown)*

**(ii)** $p/T$ = constant

$(2.6 \times 10^5) / 297 = p / (297 + 40)$ M1

$p = 2.95 \times 10^5$ Pa A0

**(b)** change in internal energy is 120 J / 25 J B1

internal energy decreases / $\underline{\Delta} U$ is negative / kinetic energy of molecules decreases M1

so temperature lower A1

*14.*

**(a)** **(i)** $N$: (total) number of <u>molecules</u> B1

**(ii)** $<c^2>$: mean square speed/velocity B1

**(b)** $pV = ⅓Nm<c^2> = NkT$

(mean) kinetic energy = ½ $m<c^2>$ C1

algebra clear leading to ½ $m<c^2> = (3/2)kT$ A1

**(c)** **(i)** *either* energy required = $(3/2) \times 1.38 \times 10^{-23} \times 1.0 \times 6.02 \times 10^{23}$ C1

= 12.5 J (*12J if 2 s.f.*) A1

*or* energy = $(3/2) \times 8.31 \times 1.0$ (C1)

= 12.5 J (A1)

**(ii)** energy is needed to push back atmosphere/do work against atmosphere M1

so total energy required is greater A1

***15.***

**(a)** the number of atoms — M1
in 12 g of carbon-12 — A1

**(b)** **(i)** amount = 3.2/40
= 0.080 mol — A1

**(ii)** $pV = nRT$
$p \times 210 \times 10^{-6} = 0.080 \times 8.31 \times 310$ — C1
$p = 9.8 \times 10^{5}$ Pa — A1
*(do not credit if T in °C not K)*

**(iii)** *either* $pV = 1/3 \times Nm <c^2>$
$N = 0.080 \times 6.02 \times 10^{23}$ $(= 4.82 \times 10^{22})$
<u>and</u> $m = 40 \times 1.66 \times 10^{-27}$ $(= 6.64 \times 10^{-26})$ — C1
$9.8 \times 10^{5} \times 210 \times 10^{-6} = 1/3 \times 4.82 \times 10^{22} \times 6.64 \times 10^{-26} \times <c^2>$ — C1
$<c^2> = 1.93 \times 10^{5}$
$c_{RMS} = 440$ m s$^{-1}$ — A1

*or* $Nm = 3.2 \times 10^{-3}$ — (C1)
$9.8 \times 10^{5} \times 210 \times 10^{-6} = 1/3 \times 3.2 \times 10^{-3} \times <c^2>$ — (C1)
$<c^2> = 1.93 \times 10^{5}$
$c_{RMS} = 440$ m s$^{-1}$ — (A1)

*or* $1/2\ m<c^2> = 3/2\ kT$ — (C1)
$1/2 \times 40 \times 1.66 \times 10^{-27} <c^2> = 3/2 \times 1.38 \times 10^{-23} \times 310$ — (C1)
$<c^2> = 1.93 \times 10^{5}$
$c_{RMS} = 440$ m s$^{-1}$ — (A1)

(*if T in °C not K award max 1/3, unless already penalised in* ***(b)(ii)***)

***16.***

**(a)** use of kelvin temperatures — B1
both values of ($V/T$) correct (11.87), $V/T$ is constant so pressure is constant — M1

(*allow use of n =1. Do not allow other values of n.*)

**(b)** **(i)** work done $= p\Delta V$
$= 4.2 \times 10^{5} \times (3.87 - 3.49) \times 10^{3} \times 10^{-6}$ — C1
$= 160$ J — A1

*(do not allow use of V instead of ΔV)*

**(ii)** increase/change in internal energy = heating of system
+ work done on system — C1
= 565 – 160
= 405 J — A1

| | |
|---|---|
| **(c)** internal energy = sum of kinetic energy and potential energy / $E_K + E_P$ | B1 |
| no intermolecular forces | M1 |
| no potential energy (so $\Delta U = \Delta E_K$) | A1 |

***17.***

| | |
|---|---|
| **(a)** obeys the equation $pV/T$ = constant | B1 |
| *(accept $pV = nRT$)* | |
| **(b) (i)** $pV = nRT$ | C1 |
| $5.0 \times 10^7 \times 3.0 \times 10^{-4} = n \times 8.31 \times 296$ giving $n = 6.1$ mol | A1 |
| **(ii)** pressure $\propto$ amount of substance | |
| loss = 0.40 / 100 × 6.1 mol = 0.0244 mol | C1 |
| $= 0.0244 \times 6.02 \times 10^{23}$ (atoms) | C1 |
| $= 1.47 \times 10^{22}$ atoms | C1 |
| rate = $(1.47 \times 10^{22}) / (35 \times 24 \times 60 \times 60)$ | |
| $= 4.9 \times 10^{15}\,s^{-1}$ | A1 |

***18.***

| | |
|---|---|
| **(a)** initially, $pV/T = (2.40 \times 10^5 \times 5.00 \times 10^{-4}) / 288 = 0.417$ | M1 |
| finally, $pV/T = (2.40 \times 10^5 \times 14.5 \times 10^{-4}) / 835 = 0.417$ | M1 |
| ideal gas because $pV/T$ is constant | A1 |
| (*allow 2 marks for two determinations of V / T and then 1 mark for V / T <u>and</u> p constant, so ideal*) | |
| **(b) (i)** work done $= p\Delta V$ | |
| $= 2.40 \times 10^5 \times (14.5 - 5.00) \times 10^{-4}$ | C1 |
| = 228 J (*ignore sign, not 2 s.f.*) | A1 |
| **(ii)** $\Delta U = q + w$ = 569 – 228 | |
| = 341 J | M1 |
| increase | A1 |

***19.***

| | |
|---|---|
| **(a)** mean speed = $1.44 \times 10^3\,m\,s^{-1}$ | A1 |
| **(b)** evidence of summing of individual squared speeds | C1 |
| mean square speed = $2.09 \times 10^6\,m^2\,s^{-2}$ | A1 |
| **(c)** root-mean-square speed = $1.45 \times 10^3\,m\,s^{-1}$ | A1 |
| *(allow ECF from **(b)** but only if arithmetic error)* | |

**20.**

| | | |
|---|---|---|
| **(a)** | obeys the equation $pV = nRT$ *or* $pV/T$ = constant | M1 |
| | all symbols explained; $T$ in kelvin/thermodynamic temperature | A1 |
| **(b) (i)** | temperature rise = 48 K | A1 |
| **(ii)** | $<c^2> \propto T$ or equivalent | C1 |
| | $<c^2> = (353/305) \times 1.9 \times 10^6$ | C1 |
| | $c_{r.m.s.} = 1480\,m\,s^{-1}$ | A1 |

**21.**

| | | |
|---|---|---|
| **(a)** | total volume of molecules negligible compared to that of containing vessel | |
| | no intermolecular forces | |
| | molecules in random motion | |
| | time of collision small compared with the time between collisions | |
| | large number of molecules | |
| | *any two* | B2 |
| **(b)** | in a real gas there is a range of velocities *or* must take the average of $v^2$ | B1 |
| **(c) (i)** | *either* $p = \frac{1}{3}\rho<c^2>$ | |
| | *or* $1.0 \times 10^5 = \frac{1}{3} \times 1.2 \times <c^2>$ | C1 |
| | $<c^2> = 2.5 \times 10^5$ | C1 |
| | $c_{r.m.s.} = 500\,m\,s^{-1}$ | A1 |
| **(ii)** | $T \propto <c^2>$ | C1 |
| | $<c^2> = 2.5 \times 10^5 \times 480/300$ | |
| | $= 4.0 \times 10^5\,m^2\,s^{-2}$ (*allow ECF from* ***(c)(i)***) | A1 |

**22.**

| | | |
|---|---|---|
| **(a) (i)** | sum of kinetic and potential energy of atoms/molecules | M1 |
| | reference to random (distribution) | A1 |
| **(ii)** | no forces (of attraction or repulsion) between molecules | B1 |
| **(b)** | $pV = NkT$ *or* $pV = nRT$ *and* $R = kN_A$, $n = N/N_A$ | B1 |
| | $^1/_3\,Nm<c^2> = NkT$ *or* $^1/_3\,m<c^2> = kT$ | B1 |
| | $<E_K> = {}^1/_2\,m<c^2>$ <u>so</u> $<E_K> = {}^3/_2\,kT$ | B1 |

(c) (i) $<E_K> = {}^3/_2 \times 1.38 \times 10^{-23} \times (273 + 12)$ C1
$= 5.9\ (5.90) \times 10^{-21}$ J A1

(*use of* $T$ = 12 K *not* $T$ = 285 K scores 0/2)

(ii) number $= (17/32) \times 6.02 \times 10^{23}$ C1
$= 3.2\ (3.20) \times 10^{23}$ A1

(iii) internal energy $= 5.9 \times 10^{-21} \times 3.2 \times 10^{23}$
$= 1900\ (1890)$ J A1

***23.***

**(a)** e.g. time of collisions negligible compared to time between collisions

no intermolecular forces (except during collisions)

random motion (of molecules)

large numbers of molecules

(total) volume of molecules negligible compared to volume of containing vessel
*or*
average/mean separation large compared with size of molecules

*any two* B2

**(b) (i)** mass = $4.0 / (6.02 \times 10^{23}) = 6.6 \times 10^{-24}$ g
*or*
mass = $4.0 \times 1.66 \times 10^{-27} \times 10^{3} = 6.6 \times 10^{-24}$ g B1

**(ii)** $\frac{3}{2}kT = \frac{1}{2}m <c^2>$ C1

$\frac{3}{2} \times 1.38 \times 10^{-23} \times 300 = \frac{1}{2} \times 6.6 \times 10^{-27} \times <c^2>$

$<c^2> = 1.88 \times 10^{6}\ (\text{m}^2\,\text{s}^{-2})$ C1

r.m.s. speed = $1.4 \times 10^{3}$ m s$^{-1}$ A1

**24.**

**(a)** **(i)** number of atoms/nuclei in 12 g of carbon-12 B1

**(ii)** amount of substance M1

containing $N_A$ (or $6.02 \times 10^{23}$) particles/molecules/atoms
*or*
which contains the same number of particles/atoms/molecules as there are atoms in 12 g of carbon-12 A1

**(b)** $pV = nRT$

$2.0 \times 10^7 \times 1.8 \times 10^4 \times 10^{-6} = n \times 8.31 \times 290$, so $n = 149$ mol or 150 mol A1

**(c)** **(i)** $V$ and $T$ constant and so pressure reduced by 5.0%
pressure = $0.95 \times 2.0 \times 10^7$ C1

*or*

calculation of new $n$ (= 142.5 mol) and correct substitution into $pV = nRT$ (C1)

pressure = $1.9 \times 10^7$ Pa A1

**(ii)** loss is $5/100 \times 150$ mol = 7.5 mol
*or*
$\Delta N = 4.52 \times 10^{24}$ C1

$t = (7.5 \times 6.02 \times 10^{23}) / 1.5 \times 10^{19}$
*or*
$t = 4.52 \times 10^{24} / 1.5 \times 10^{19}$ C1

$= 3.0 \times 10^5$ s A1

**25.**

**(a)** **(i)** number of moles/amount of substance B1

**(ii)** kelvin temperature/absolute temperature/thermodynamic temperature B1

**(b)** $pV = nRT$

$4.9 \times 10^5 \times 2.4 \times 10^3 \times 10^{-6} = n \times 8.31 \times 373$ B1

$n = 0.38$ (mol) C1

number of molecules or $N = 0.38 \times 6.02 \times 10^{23} = 2.3 \times 10^{23}$ A1 *or*

$pV = NkT$ (C1)

$4.9 \times 10^5 \times 2.4 \times 10^3 \times 10^{-6} = N \times 1.38 \times 10^{-23} \times 373$ (M1)

number of molecules or $N = 2.3 \times 10^{23}$ (A1)

**(c)** volume occupied by one molecule $= (2.4 \times 10^3) / (2.3 \times 10^{23})$ C1

$= 1.04 \times 10^{-20}$ cm$^3$

mean spacing $= (1.04 \times 10^{-20})^{1/3}$ C1

$= 2.2 \times 10^{-7}$ cm (*allow 1 s.f.*) A1

(*allow other dimensionally correct methods e.g.* $V = (4/3)\pi r^3$)

***26.***

| | | |
|---|---|---|
| 4(a) | random/haphazard | **B1** |
| | constant velocity **or** speed in a straight line between collisions<br>**or**<br>distribution of speeds/different directions | **B1** |
| 4(b) | (small) specks of light/bright specks/pollen grains/dust particles/smoke particles | **M1** |
| | moving haphazardly/randomly/jerky/in a zigzag fashion | **A1** |
| 4(c)(i) | $pV = \frac{1}{3} Nm\langle c^2\rangle$<br>$1.05 \times 10^5 \times 0.0240 = \frac{1}{3} \times 4.00 \times 10^{-3} \times \langle c^2\rangle$ | **C1** |
| | $\langle c^2\rangle = 1.89 \times 10^6$ | **C1** |
| | **or** | |
| | $\frac{1}{2} m\langle c^2\rangle = (3/2)\,kT$<br>$0.5 \times (4.00 \times 10^{-3} / 6.02 \times 10^{23}) \times \langle c^2\rangle = 1.5 \times 1.38 \times 10^{-23} \times 300$ | **(C1)** |
| | $\langle c^2\rangle = 1.87 \times 10^6$ | **(C1)** |
| | **or** | |
| | $nRT = \frac{1}{3} Nm\langle c^2\rangle$<br>$1.00 \times 8.31 \times 300 = \frac{1}{3} \times 4.00 \times 10^{-3} \times \langle c^2\rangle$ | **(C1)** |
| | $\langle c^2\rangle = 1.87 \times 10^6$ | **(C1)** |
| | $c_{r.m.s.} = 1.37 \times 10^3$ m s$^{-1}$ | **A1** |
| 4(c)(ii) | $\langle c^2\rangle \propto T$ | **C1** |
| | $\langle c^2\rangle$ at 177 °C $= 1.89 \times 10^6 \times (450 / 300)$ | **C1** |
| | $c_{r.m.s.}$ at 177 °C $= 1.68 \times 10^3$ m s$^{-1}$ | **A1** |

## *27.*

| | | |
|---|---|---|
| 2(a)(i) | mean/average square speed/velocity | **B1** |
| 2(a)(ii) | $pV = NkT$ **or** $pV = nRT$ | **B1** |
| | $\rho = Nm / V$<br>**or**<br>$\rho = nN_A m / V$ **and** $k = nR / N$ | **B1** |
| | $E_K = ½\, m\langle c^2\rangle$ with algebra to $(3 / 2)kT$ | **B1** |
| 2(b)(i) | no (external) work done **or** $\Delta U = q$ **or** $w = 0$ | **B1** |
| | $q = N_A \times (3 / 2)k \times 1.0$ | **M1** |
| | $N_A k = R$ so $q = (3 / 2)R$ | **A1** |
| 2(b)(ii) | specific heat capacity $= \{(3 / 2) \times R\} / 0.028$ | **C1** |
| | $= 450\ \text{J kg}^{-1}\,\text{K}^{-1}$ | **A1** |

## *28.*

| | | |
|---|---|---|
| 2(a) | $pV = nRT$<br>$T = (5.60 \times 10^5 \times 3.80 \times 10^{-2}) / (5.12 \times 8.31)$ | **C1** |
| | $T =$ **500 K** | **A1** |
| 2(b)(i) | $V / T$ is constant<br>$V = (3.80 \times 10^4) \times (500 + 125) / 500$ | **C1** |
| | $V = \mathbf{4.75 \times 10^4}\ \text{cm}^3$ | **A1** |
| 2(b)(ii) | (for ideal gas,) change in internal energy is change in (total) kinetic energy (of molecules) | **B1** |
| | $\Delta U = 3 / 2 \times 1.38 \times 10^{-23} \times 125 \times 5.12 \times 6.02 \times 10^{23}$ | **C1** |
| | = **7980** J | **A1** |
| 2(c)(i) | $w = p\Delta V$<br>$= 5.60 \times 10^5 \times (4.75 - 3.80) \times 10^{-2}$ | **C1** |
| | = **5320** J | **A1** |
| 2(c)(ii) | total = 7980 + 5320<br>= **13300** J | **A1** |

## *29.*

| | | |
|---|---|---|
| 2(a) | no intermolecular forces (so no potential energy) | **B1** |
| 2(b)(i) | mean square speed (of molecule(s)) | **B1** |
| 2(b)(ii) | kelvin/thermodynamic/absolute <u>temperature</u> | **B1** |
| 2(c)(i)1. | $pV = NkT$ | **C1** |
| | $4.7 \times 10^{-2} \times 2.6 \times 10^5 = N \times 1.38 \times 10^{-23} \times 446$ | **C1** |
| | **or** | |
| | $pV = nRT$ **and** $N = nN_A$<br>$4.7 \times 10^{-2} \times 2.6 \times 10^5 = n \times 8.31 \times 446$<br>$n = 3.3$ (mol) | **(C1)** |
| | $N = 3.3 \times 6.02 \times 10^{23}$ | **(C1)** |
| | $N = 2.0 \times 10^{24}$ | **A1** |
| 2(c)(i)2. | average increase $= 2900 / (2.0 \times 10^{24})$<br>$= 1.5 \times 10^{-21}$ J | **A1** |

| | | |
|---|---|---|
| 2(c)(ii) | $\Delta E_K = (3/2)k\,(\Delta)T$<br>$1.5 \times 10^{-21} = (3/2) \times 1.38 \times 10^{-23} \times (\Delta)T$ | C1 |
| | $(\Delta)T$ in range 70–72 K | C1 |
| | $T = 173 + 273 + 70$<br>$= 520$ K | A1 |

## 30.

| | | |
|---|---|---|
| 2(a) | gas that obeys equation $pV$ = constant $\times T$ | M1 |
| | symbols $p, V$ and $T$ explained | A1 |
| 2(b)(i) | $pV = \frac{1}{3} Nm\langle c^2\rangle$ **and** $M = Nm$<br>(and so) $p = \frac{1}{3}\rho\langle c^2\rangle$ | C1 |
| | $2.12 \times 10^7 = \frac{1}{3} \times [3.20/(1.84 \times 10^{-2})] \times \langle c^2\rangle$ | C1 |
| | $c_{r.m.s.} = 605\,m\,s^{-1}$ | A1 |
| 2(b)(ii) | **1.** $pV = nRT$ **and** $T = (22 + 273)$K | C1 |
| | $n = (2.12 \times 10^7 \times 1.84 \times 10^{-2})/(8.31 \times 295)$<br>$= 159$ mol | A1 |
| | **2.** mass $= 3.20/(159 \times 6.02 \times 10^{23})$<br>**or**<br>mass $= [2 \times (3/2) \times 1.38 \times 10^{-23} \times 295]/605^2$ | C1 |
| | mass $= 3.34 \times 10^{-26}$ kg | A1 |
| 2(c) | $A = (3.34 \times 10^{-26})/(1.66 \times 10^{-27})$<br>$= 20$ | A1 |

## 31.

| | | |
|---|---|---|
| 2(a)(i) | gas obeys formula $pV/T$ = constant | M1 |
| | symbols $V$ and $T$ explained | A1 |
| 2(a)(ii) | mean-square-speed (of atoms / molecules) | B1 |
| 2(b)(i) | use of $T = 393$ | C1 |
| | $pV = nRT$ | C1 |
| | $2.4 \times 10^5 \times 6.8 \times 10^{-3} = n \times 8.31 \times 393$<br>and $N = n \times 6.02 \times 10^{23} = 3.0 \times 10^{23}$ | A1 |
| | or<br>$pV = NkT$ | (C1) |
| | $2.4 \times 10^5 \times 6.8 \times 10^{-3} = N \times 1.38 \times 10^{-23} \times 393$<br>hence $N = 3.0 \times 10^{23}$ | (A1) |
| 2(b)(ii) | volume of one atom $= 4/3\pi r^3$ | C1 |
| | volume occupied $= 3.0 \times 10^{23} \times 4/3 \times \pi \times (3.2 \times 10^{-11})^3$<br>$= 4 \times 10^{-8}\,m^3$ | A1 |
| 2(b)(iii) | assumption: volume of atoms negligible compared to volume of container / cylinder | B1 |
| | $4 \times 10^{-8}\,(m^3) \ll 6.8 \times 10^{-3}\,(m^3)$ so yes | B1 |

## *32.*

| | | |
|---|---|---|
| 2(a)(i) | $pV = nRT$ | **C1** |
| | $n = (3.0 \times 10^5 \times 210 \times 10^{-6}) / (8.31 \times 270)$ | **C1** |
| | $= 0.028$ mol | **A1** |
| 2(a)(ii) | $V \propto T$ **or** $T = pV / nR$ with value of $n$ from **(i)** | **C1** |
| | $T = (140 / 210) \times 270$<br>**or**<br>$T = (3.0 \times 10^5 \times 140 \times 10^{-6}) / (8.31 \times 0.028)$<br>$= 180$ K | **A1** |
| 2(a)(iii) | $W = p\Delta V$<br>$= 3.0 \times 10^5 \times (210 - 140) \times 10^{-6}$ | **C1** |
| | $= 21$ J | **A1** |
| 2(b) | $\Delta U = w + q$ | **C1** |
| | $= 21 - 53$ | **C1** |
| | **or** | |
| | $\Delta U = (nN_A) \times (3/2)k\Delta T$ | **(C1)** |
| | $= (0.0281 \times 6.02 \times 10^{23}) \times (3/2) \times 1.38 \times 10^{-23} \times (180 - 270)$ | **(C1)** |
| | **or** | |
| | $\Delta U = (3/2)nR\Delta T$ | **(C1)** |
| | $= (3/2) \times 0.0281 \times 8.31 \times (180 - 270)$ | **(C1)** |
| | $\Delta U = (-)32$ J | **A1** |

## *33.*

| | | |
|---|---|---|
| 2(a) | (total volume of molecules is) negligible | **M1** |
| | compared with volume occupied by the gas | **A1** |
| 2(b)(i) | $pV = NkT$ | **C1** |
| | $4.60 \times 10^5 \times 2.40 \times 10^{-2} = N \times 1.38 \times 10^{-23} \times (273 + 23)$ | **C1** |
| | **or** | |
| | $pV = nRT$ | **(C1)** |
| | $4.60 \times 10^5 \times 2.40 \times 10^{-2} = n \times 8.31 \times (273 + 23)$<br>$n = 4.49$ (mol)<br>$N = nN_A$<br>$= 4.49 \times 6.02 \times 10^{23}$ | **(C1)** |
| | $N = 2.7 \times 10^{24}$ | **A1** |

| | | |
|---|---|---|
| 2(b)(ii) | volume of one atom $= d^3$ ($= 2.7 \times 10^{-29}$ m$^3$) | **C1** |
| | volume of all atoms $= 2.7 \times 10^{-29} \times 2.7 \times 10^{24}$ | **C1** |
| | $= 7 \times 10^{-5}$ m$^3$ | **A1** |
| | **or** | |
| | volume of one atom $= (4/3)\pi r^3$ ($= 1.41 \times 10^{-29}$ m$^3$) | **(C1)** |
| | volume of all atoms $= 2.7 \times 10^{24} \times 1.41 \times 10^{-29}$ | **(C1)** |
| | $= 4 \times 10^{-5}$ m$^3$ | **(A1)** |
| 2(c) | numerical comparison between answer to **(b)(ii)** and $2.4 \times 10^{-2}$ (m$^3$) showing **(b)(ii)** is <u>much</u> less than $2.4 \times 10^{-2}$ (m$^3$) | **B1** |

## 34.

| | | |
|---|---|---|
| 2(a)(i) | specks of light moving haphazardly | **B1** |
| 2(a)(ii) | (gas) molecules collide with (smoke) particles<br>**or**<br>random motion of the (gas) molecules | **M1** |
| | <u>causes</u> the (haphazard) motion of the smoke particles<br>**or**<br><u>causes</u> the smoke particles to change direction | **A1** |
| 2(b)(i) | $pV = nRT$ | **C1** |
| | $n = (3.51 \times 10^5 \times 2.40 \times 10^{-3}) / (8.31 \times 290)$<br>**or**<br>$n = (3.75 \times 10^5 \times 2.40 \times 10^{-3}) / (8.31 \times 310)$ | **C1** |
| | **or** | |
| | $pV = NkT$ | **(C1)** |
| | $n = (3.51 \times 10^5 \times 2.40 \times 10^{-3}) / (1.38 \times 10^{-23} \times 6.02 \times 10^{23} \times 290)$<br>**or**<br>$n = (3.75 \times 10^5 \times 2.40 \times 10^{-3}) / (1.38 \times 10^{-23} \times 6.02 \times 10^{23} \times 310)$ | **(C1)** |
| | $n = 0.350$ mol **or** 0.349 mol | **A1** |
| 2(b)(ii) | energy transfer = (0.349 **or** 0.35) $\times 12.5 \times (310 - 290)$ | **C1** |
| | = 87.3 J **or** 87.5 J | **A1** |
| 2(c)(i) | zero | **A1** |
| 2(c)(ii) | 87.3 J **or** 87.5 J | **A1** |
| | increase | **B1** |

## 35.

| | | |
|---|---|---|
| 2(a) | $n = 110 / 0.032$ or $110000 / 32$ or 3440 | **C1** |
| | $pV = nRT$ | **C1** |
| | $T = (1.0 \times 10^5 \times 85) / (8.31 \times (110 / 0.032)) = 300$ K | **A1** |
| 2(b) | $E = mc\Delta\theta$<br>$= 110 \times 0.66 \times 50$ | **C1** |
| | = **3600** J | **A1** |
| 2(c) | Any 3 from:<br>• molecule collides with wall<br>• momentum of molecule changes during collision (with wall)<br>• force on molecule so force on wall<br>• many forces act over surface area of container exerting a pressure | **B3** |

| | | |
|---|---|---|
| 2(d) | $KE \propto T$<br>$v \propto \sqrt{T}$ | C1 |
| | ratio $= \sqrt{(350 / 300)}$<br>$= \mathbf{1.1}$ | A1 |

## 36.

| | | |
|---|---|---|
| 2(a) | $\rho$: $Nm / V$ | B1 |
| | ⅓: molecules move in three dimensions (not one)<br>so ⅓ in any (one) direction | B1 |
| | $<c^2>$: molecules have different speeds so take average | M1 |
| | of $(\text{speed})^2$ | A1 |
| 2(b) | $pV = NkT$ | C1 |
| | $N = (3.0 \times 10^5 \times 6.0 \times 10^{-3}) / (1.38 \times 10^{-23} \times 290)$ | C1 |
| | $= 4.5 \times 10^{23}$ | C1 |
| | mass $= 20.7 / (4.5 \times 10^{23})$<br>$= 4.6 \times 10^{-23}$ g | A1 |

## 37.

| | | |
|---|---|---|
| 2(a) | sum of potential energy and kinetic energy (of particles) | B1 |
| | (total) energy of random motion of particles | B1 |
| 2(b)(i) | $pV = nRT$ | C1 |
| | $2.60 \times 10^5 \times 2.30 \times 10^{-3} = n \times 8.31 \times 180$<br>$n = 0.400$ mol | A1 |
| 2(b)(ii) | $(2.30 \times 10^{-3}) / 180 = (3.80 \times 10^{-3}) / T$<br>**or**<br>$2.60 \times 10^5 \times 3.80 \times 10^{-3} = 0.400 \times 8.31 \times T$ | C1 |
| | $T = 297$ K | A1 |
| 2(c)(i) | $\Delta W = p\Delta V$<br>$= 2.60 \times 10^5 \times (2.30 - 3.80) \times 10^{-3}$ | C1 |
| | $= (-)390$ J | A1 |
| | negative because work is done by gas<br>**or**<br>negative because work is done against atmospheric pressure<br>**or**<br>negative because volume of gas increases | B1 |
| 2(c)(ii) | $\Delta U = (980 - 390)$<br>$= 590$ J | A1 |

## *38.*

| | | |
|---|---|---|
| 2(a)(i) | $pV = NkT$ or $pV = nRT$ <u>and</u> $N = nN_A$<br><br>$N = \dfrac{2.3\times10^5\times3.5\times10^{-3}}{1.38\times10^{-23}\times294}$ | **C1** |
| | = **2.0 × 10²³** | **A1** |
| 2(a)(ii) | $pV = \dfrac{1}{3}Nmc^2$<br><br>$c^2 = \dfrac{3\times2.3\times10^5\times3.5\times10^{-3}}{2.0\times10^{23}\times40\times1.66\times10^{-27}}$<br><br>= 182 000<br><br>r.m.s. speed = **430** m s⁻¹ | **C1** |
| | or<br><br>$\frac{1}{2}mc^2 = \frac{3}{2}kT$ | **A1** |
| | $c^2 = \dfrac{3\times1.38\times10^{-23}\times294}{40\times1.66\times10^{-27}}$<br><br>= 183 000 | **(C1)** |
| | r.m.s.speed = 430 m s⁻¹ | **(A1)** |
| 2(b) | $c^2 = \dfrac{3\times2.0\times10^{23}\times1.38\times10^{-23}\times(294+84)}{2.0\times10^{23}\times40\times1.66\times10^{-27}}$<br><br>$c^2 = 236\,000$<br><br>$c = 485$ | **C1** |
| | $ratio\left(=\dfrac{485}{430}\right) = 1.1$ | **A1** |
| | OR<br>$v \propto \sqrt{T}$ or $v^2 \propto T$ | **(C1)** |
| | $ratio = \sqrt{\dfrac{273+21+84}{273+21}}$ or $\sqrt{\dfrac{378}{294}}$<br><br>ratio = **1.1** | **(A1)** |

## *39.*

| | | |
|---|---|---|
| 2(a) | $pV = NkT$ | **C1** |
| | $N = (1.8 \times 10^{-3} \times 3.3 \times 10^5) / (1.38 \times 10^{-23} \times 310) = 1.4 \times 10^{23}$ | **A1** |
| | **or** | |
| | $pV = nRT$ **and** $nN_A = N$ | **(C1)** |
| | $N = (1.8 \times 10^{-3} \times 3.3 \times 10^5 \times 6.02 \times 10^{23}) / (8.31 \times 310) = 1.4 \times 10^{23}$ | **(A1)** |
| 2(b) | speed of molecule decreases on impact with moving piston | **B1** |
| | mean square speed (directly) proportional to (thermodynamic) temperature<br>**or**<br>mean square speed (directly) proportional to kinetic energy (of molecules)<br>**or**<br>kinetic energy (of molecules) (directly) proportional to (thermodynamic) temperature | **B1** |
| | kinetic energy (of molecules) decreases (so temperature decreases) | **B1** |

| | | |
|---|---|---|
| 2(c)(i) | $\Delta U = 3/2 \times k \times \Delta T \times N$ | C1 |
| | $= 3/2 \times 1.38 \times 10^{-23} \times (288 - 310) \times 1.4 \times 10^{23}$ | C1 |
| | $= - 64$ J | A1 |
| 2(c)(ii) | decrease in internal energy is less than work done by gas | M1 |
| | (thermal energy is) transferred <u>to</u> the gas (during the expansion) | A1 |

## 40.

| | | |
|---|---|---|
| 2(a) | $pV = nRT$ | C1 |
| | $pV = nRT$ **and** $N = nN_A$<br>**or**<br>$pV = NkT$ | C1 |
| | $3.1 \times 10^{-3} \times 8.5 \times 10^5 = (N \times 290 \times 8.31) / (6.02 \times 10^{23})$<br>so $N = 6.6 \times 10^{23}$<br><br>**or**<br><br>$3.1 \times 10^{-3} \times 8.5 \times 10^5 = N \times 1.38 \times 10^{-23} \times 290$<br>so $N = 6.6 \times 10^{23}$ | A1 |
| 2(b)(i) | $(3.1 \times 10^{-3} \times 8.5 \times 10^5) / 290 = (6.3 \times 10^{-3} \times 2.7 \times 10^5) / T$<br>so $T = 190$ K<br><br>**or**<br><br>$6.3 \times 10^{-3} \times 2.7 \times 10^5 = 6.6 \times 10^{23} \times 1.38 \times 10^{-23} \times T$<br>so $T = 190$ K | A1 |
| 2(b)(ii) | $\Delta U = 3/2 \times k \times \Delta T \times N$ | C1 |
| | $= 3/2 \times 1.38 \times 10^{-23} \times (190 - 290) \times 6.6 \times 10^{23}$ | C1 |
| | $= -1400$ J | A1 |
| 2(c) | $\Delta U = q + w$ | M1 |
| | $q = 0$ so $\Delta U = w$ | A1 |

## 41.

| | | |
|---|---|---|
| 3(a)(i) | no loss of <u>kinetic</u> energy | B1 |
| 3(a)(ii) | • <u>molecules</u> have negligible volume (compared with gas/container)<br>• no forces between <u>molecules</u> (except during collisions)<br>• <u>molecules</u> are in random motion<br>• collisions are instantaneous<br>*Any two points, 1 mark each* | B2 |
| 3(b)(i) | $2mu$ | A1 |
| 3(b)(ii) | $2L / u$ | A1 |
| 3(b)(iii) | force = change in momentum / time = $2mu / (2L / u)$<br>$= mu^2 / L$ | A1 |
| 3(b)(iv) | pressure = force / area = $(mu^2 / L) / L^2$<br>$= mu^2 / L^3$ | A1 |
| 3(c) | $pV = NkT$ | C1 |
| | $NkT = \frac{1}{3}Nm\langle c^2\rangle$ leading to $\frac{1}{2}m\langle c^2\rangle = (3/2)kT$ <u>and</u> $\frac{1}{2}m\langle c^2\rangle = E_K$ | A1 |
| 3(d) | $\frac{1}{2} \times 3.34 \times 10^{-27} \times \langle c^2\rangle = (3/2) \times 1.38 \times 10^{-23} \times (25 + 273)$ | C1 |
| | r.m.s. speed $= 1.9 \times 10^3$ m s$^{-1}$ | A1 |

## 42.

| | | |
|---|---|---|
| 3(a) | $p$ = pressure (of gas), $V$ = volume (of gas) **and** $k$ = Boltzmann constant | **B1** |
| | $N$ = number of molecules | **B1** |
| | $T$ = thermodynamic temperature | **B1** |
| 3(b) | ($pV = NkT$ and $pV = ⅓Nm\langle c^2\rangle$ leading to) $NkT = ⅓Nm\langle c^2\rangle$ | **M1** |
| | algebra leading to $(3/2)kT = ½m\langle c^2\rangle$ **and** use of $½m\langle c^2\rangle = E_K$ leading to $(3/2)kT = E_K$ | **A1** |
| 3(c)(i) | $T = 296$ K | **C1** |
| | $½m\langle c^2\rangle = (3/2)kT$<br>$½ \times 5.31 \times 10^{-26} \times u^2 = (3/2) \times 1.38 \times 10^{-23} \times 296$ | **C1** |
| | $u = 480$ m s$^{-1}$ | **A1** |
| 3(c)(ii) | line passing through ($P$, $u$) | **B1** |
| | horizontal straight line | **B1** |

## 43.

| | | |
|---|---|---|
| 2(a)(i) | (gas that obeys) $pV \propto T$ (for all values of $p$, $V$ and $T$) | **M1** |
| | where $T$ is thermodynamic temperature | **A1** |
| 2(a)(ii) | temperature = –273.15 °C | **A1** |
| 2(b)(i) | $pV = NkT$ | **C1** |
| | $N = (1.37 \times 10^5 \times 0.640) / (1.38 \times 10^{-23} \times (227 + 273))$ | **C1** |
| | $= 1.27 \times 10^{25}$ | **A1** |
| 2(b)(ii) | mass $= 0.0424 / (1.27 \times 10^{25})$<br>$= 3.34 \times 10^{-27}$ kg | **A1** |
| 2(b)(iii) | $½m\langle c^2\rangle = (3/2)kT$ | **C1** |
| | $3.34 \times 10^{-27} \times v^2 = 3 \times 1.38 \times 10^{-23} \times 500$ | **C1** |
| | $v = 2490$ m s$^{-1}$ | **A1** |
| | **or** | |
| | $pV = ⅓(Nm)\langle c^2\rangle$ <u>and</u> $Nm$ = mass of gas | **(C1)** |
| | $0.0424 \times v^2 = 3 \times 1.37 \times 10^5 \times 0.640$ | **(C1)** |
| | $v = 2490$ m s$^{-1}$ | **(A1)** |
| 2(c) | sketch: line from (0, 0) to (500, $v$) | **B1** |
| | line with decreasing positive gradient throughout | **B1** |